

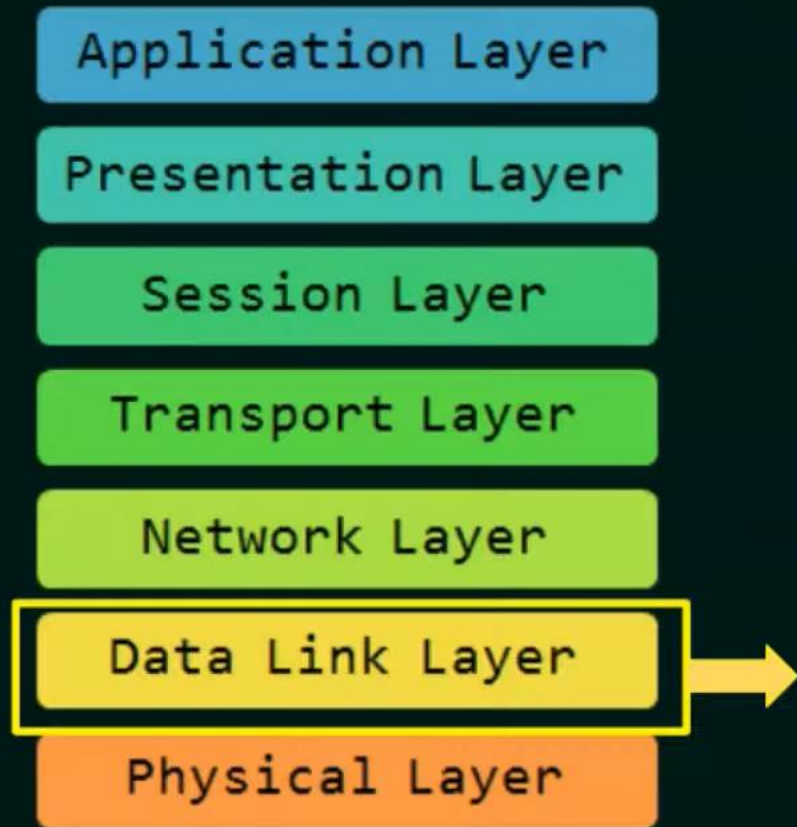


COMPUTER NETWORKS

A Bottom up approach 

Flow Control

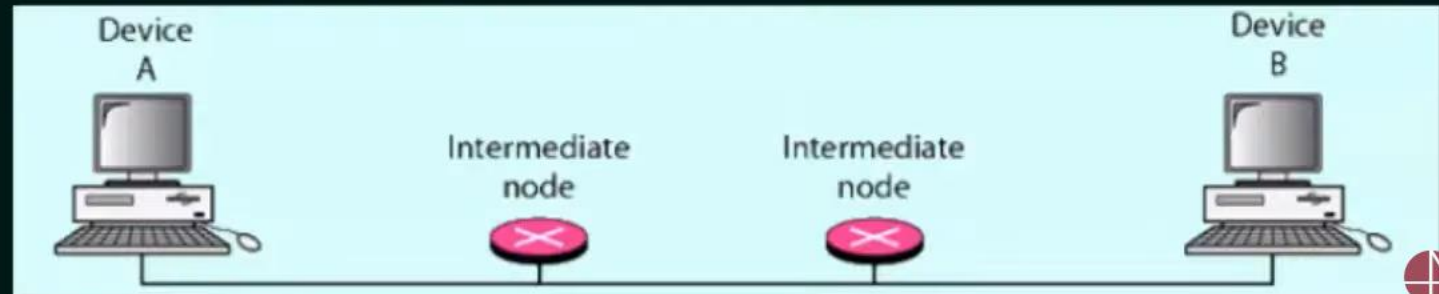
LINK LAYER SERVICES



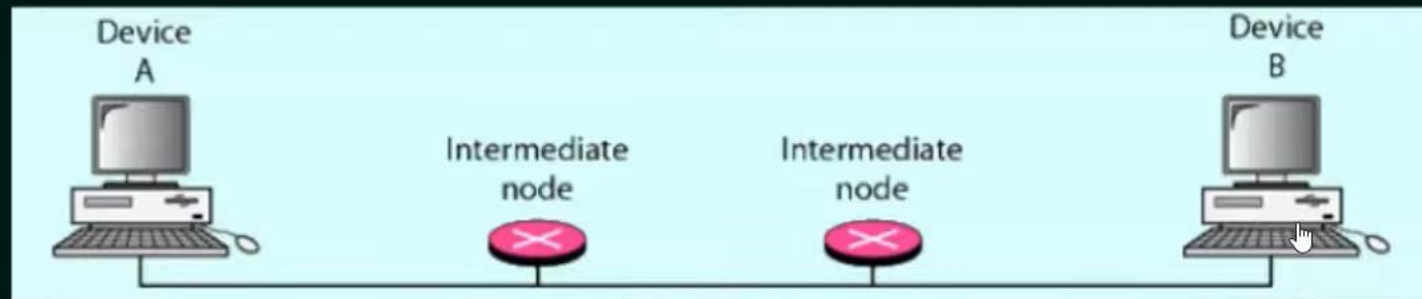
It is responsible for moving data(frames) from one node to another node.

SERVICES PROVIDED BY DATA LINK LAYER

- ★ Framing.
- ★ Error Control.
- ★ Flow Control.
- ★ Physical Addressing.
- ★ Access Control.



FLOW CONTROL

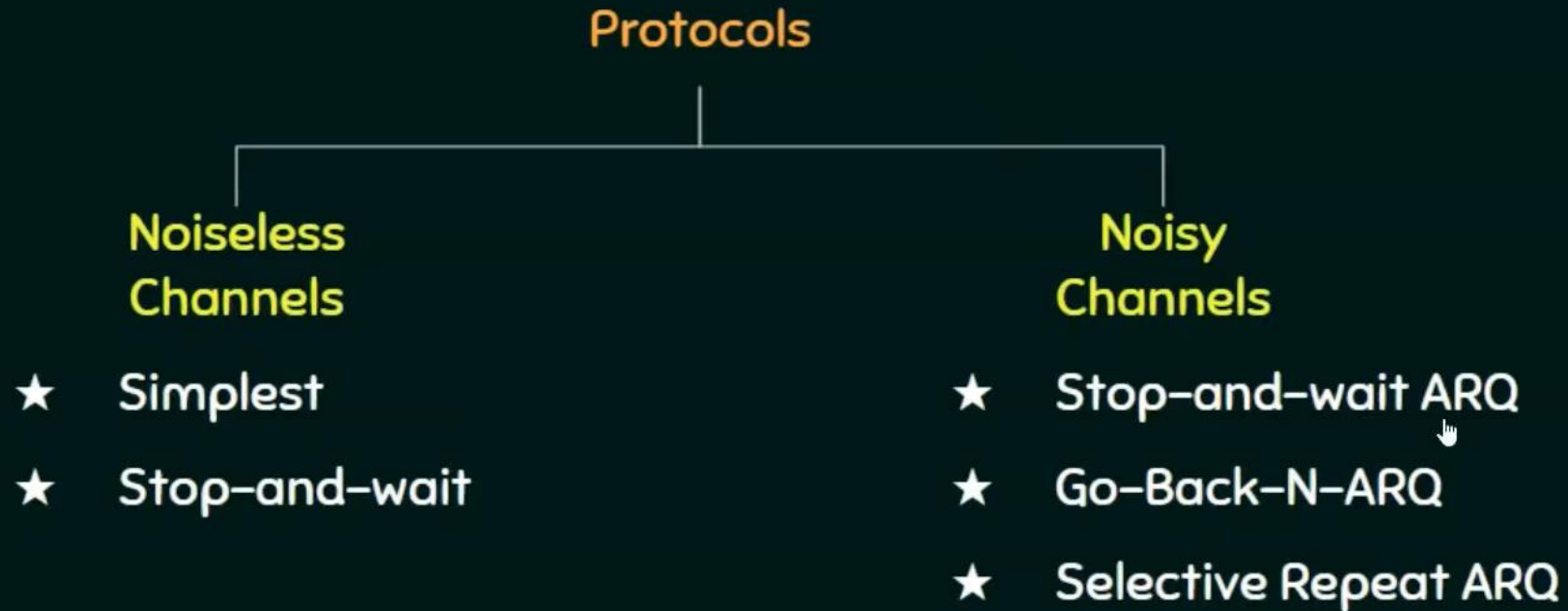


FLOW CONTROL

- ★ Speed matching mechanism.
- ★ Flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- ★ Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- ★ Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- ★ Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.



FLOW CONTROL – PROTOCOLS



STOP-AND-WAIT PROTOCOL

- ★ Stop – and – Wait protocol is data link layer protocol for transmission of frames over noiseless channels.
- ★ It provides unidirectional data transmission with flow control facilities but without error control facilities.
- ★ The idea of stop-and-wait protocol is straightforward.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.



PRIMITIVES OF STOP-AND-WAIT PROTOCOL

Sender side

Rule 1 : Send one data packet at a time.

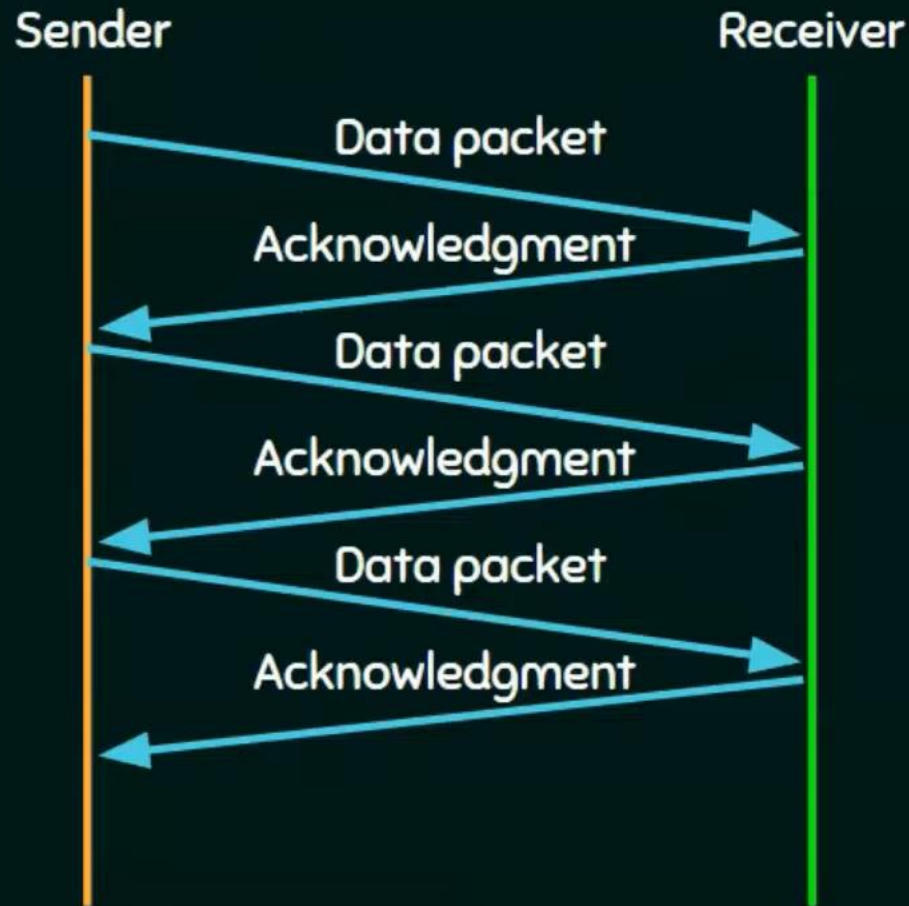
Rule 2 : Send the next packet only after receiving ACK for the previous.

Receiver side

Rule 1 : Receive and consume data packet.

Rule 2 : After consuming packet, ACK need to be sent (Flow Control).

STOP-AND-WAIT PROTOCOL



PROBLEMS OF STOP-AND-WAIT PROTOCOL

1. Problems due to lost data.

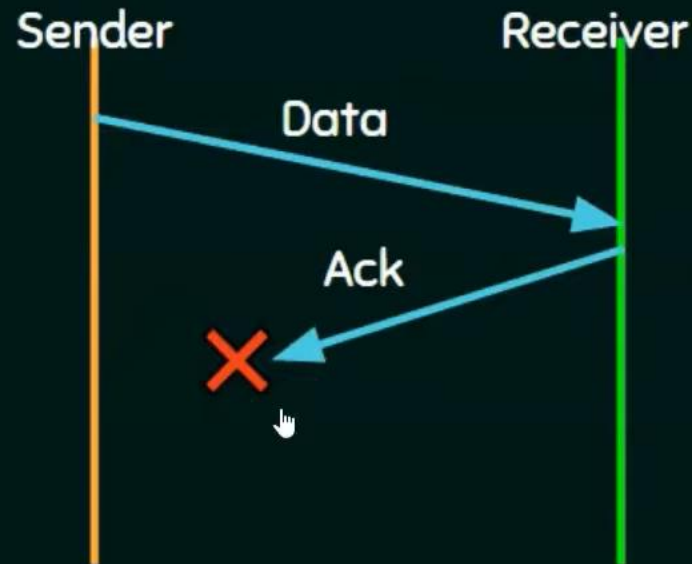
- ★ Sender waits for ack for an infinite amount of time.
- ★ Receiver waits for data an infinite amount of time.



PROBLEMS OF STOP-AND-WAIT PROTOCOL

2. Problems due to lost ACK.

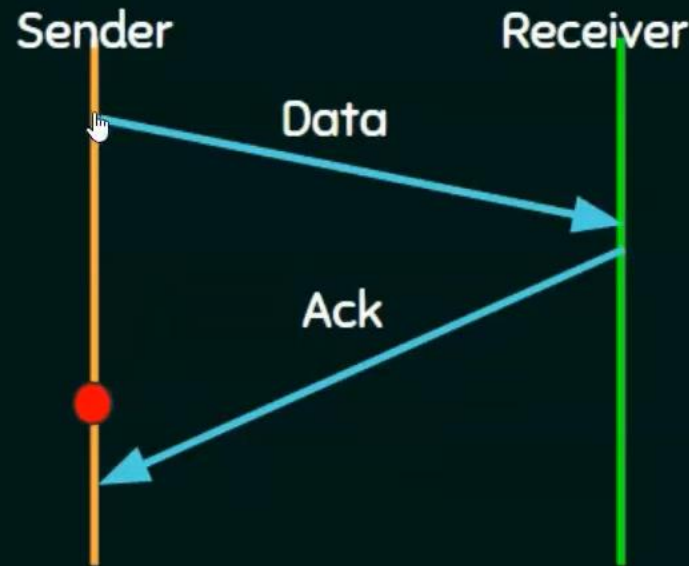
- ★ Sender waits for an infinite amount of time for ack.



PROBLEMS OF STOP-AND-WAIT PROTOCOL

3. Problems due to delayed ACK/data.

- ★ After timeout on sender side, a delayed ack might be wrongly considered as ack of some other data packet.

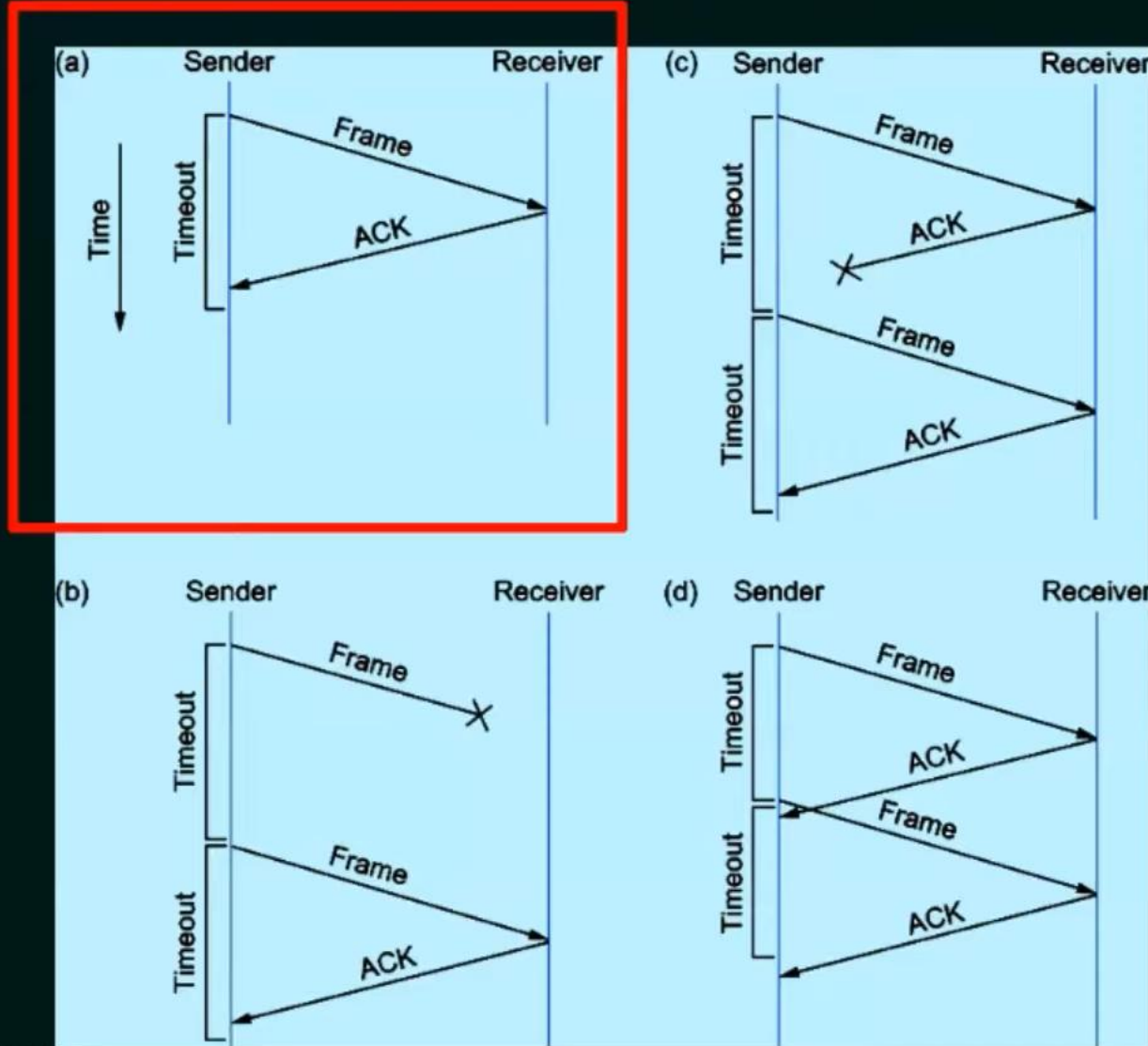


STOP-AND-WAIT ARQ PROTOCOL

- ★ Idea of stop-and-wait protocol is straightforward.
- ★ After transmitting one frame, the sender waits for an acknowledgement before transmitting the next frame.
- ★ If the acknowledgement does not arrive after a certain period of time, the sender times out and retransmits the original frame.
- ★ Stop-and-Wait ARQ = Stop-and-Wait + Timeout Timer + Sequence number

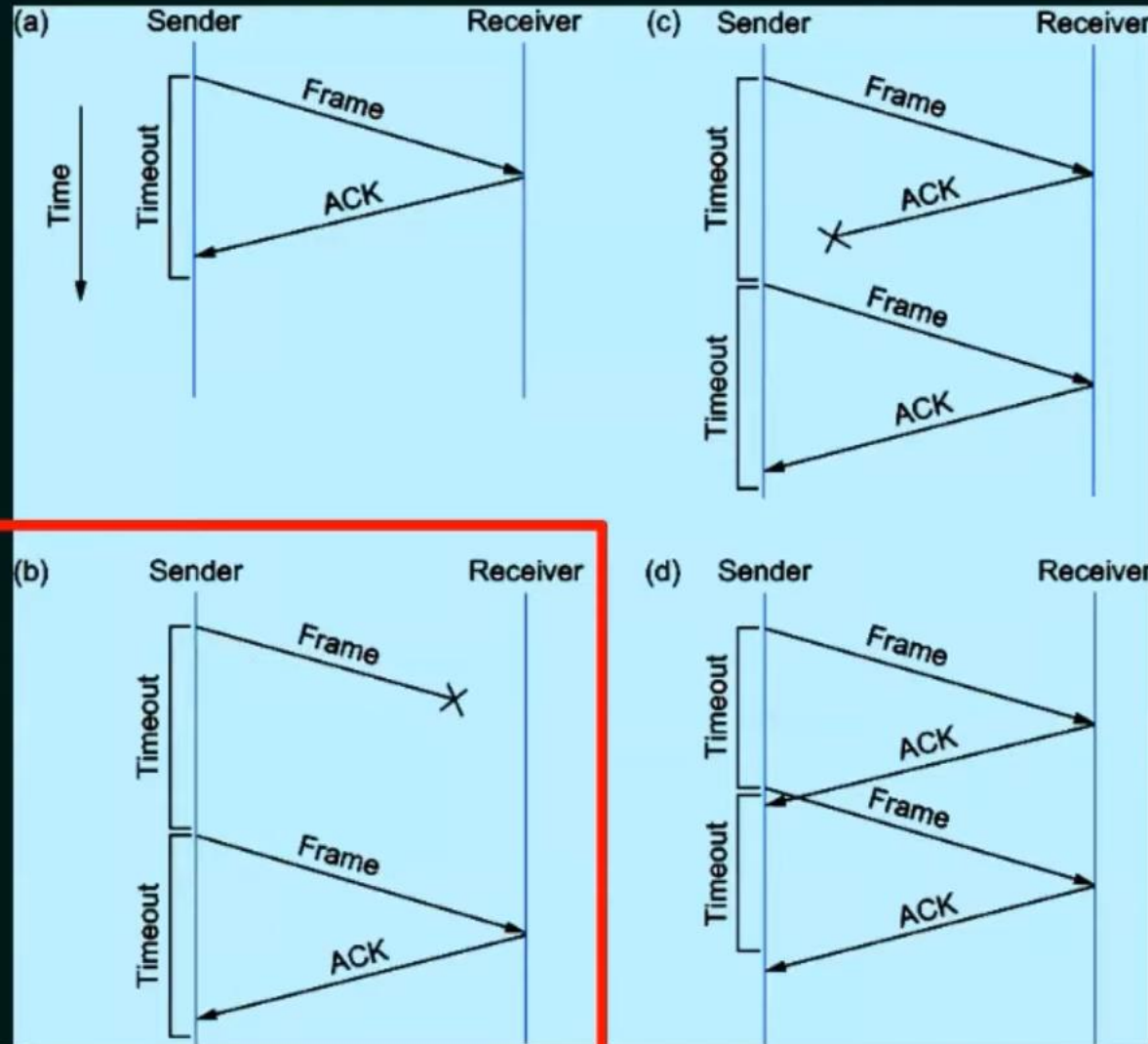


STOP-AND-WAIT ARQ PROTOCOL



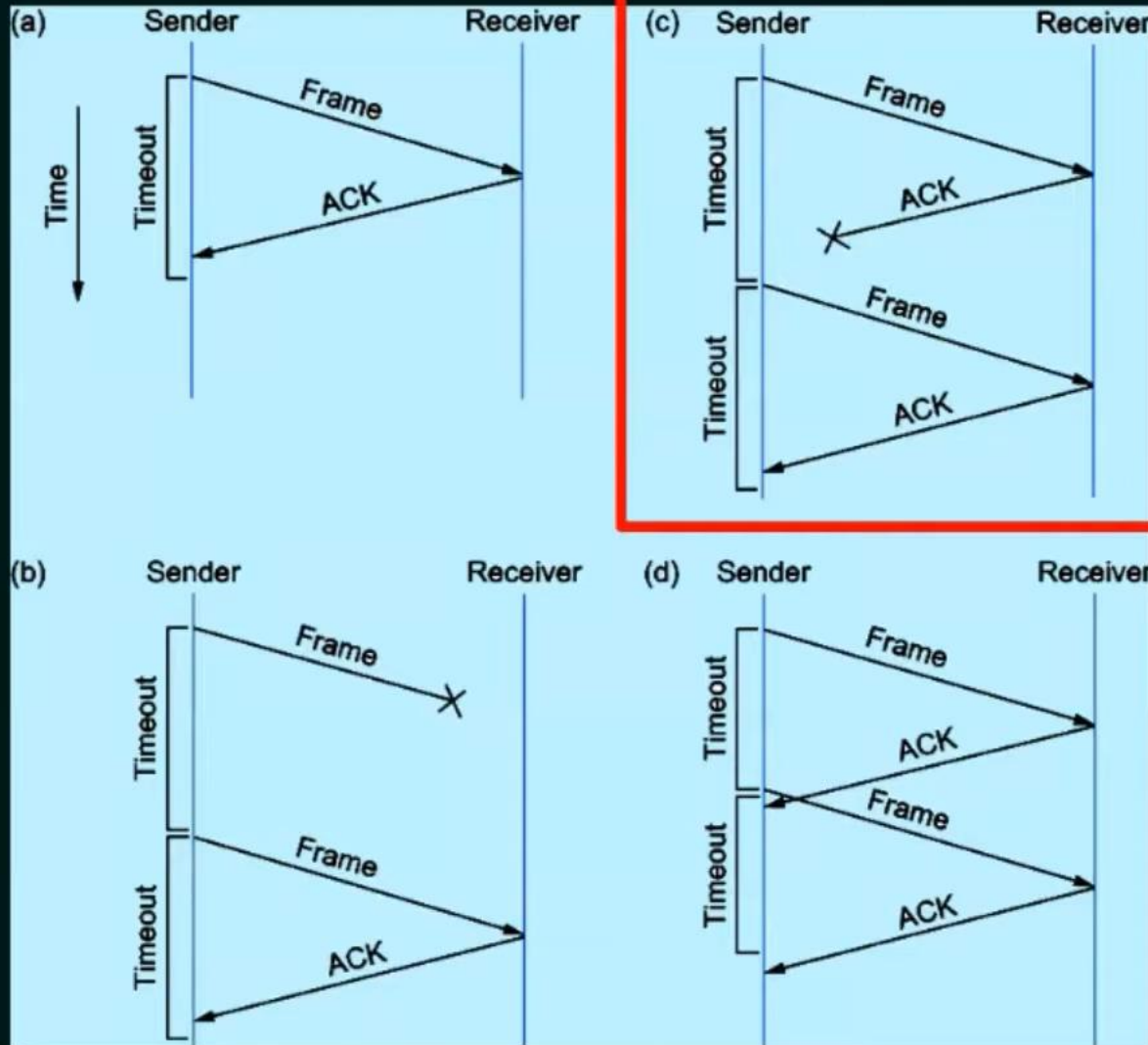
(a) The ACK is received before the timer expires;

STOP-AND-WAIT ARQ PROTOCOL



(b) The original frame is lost

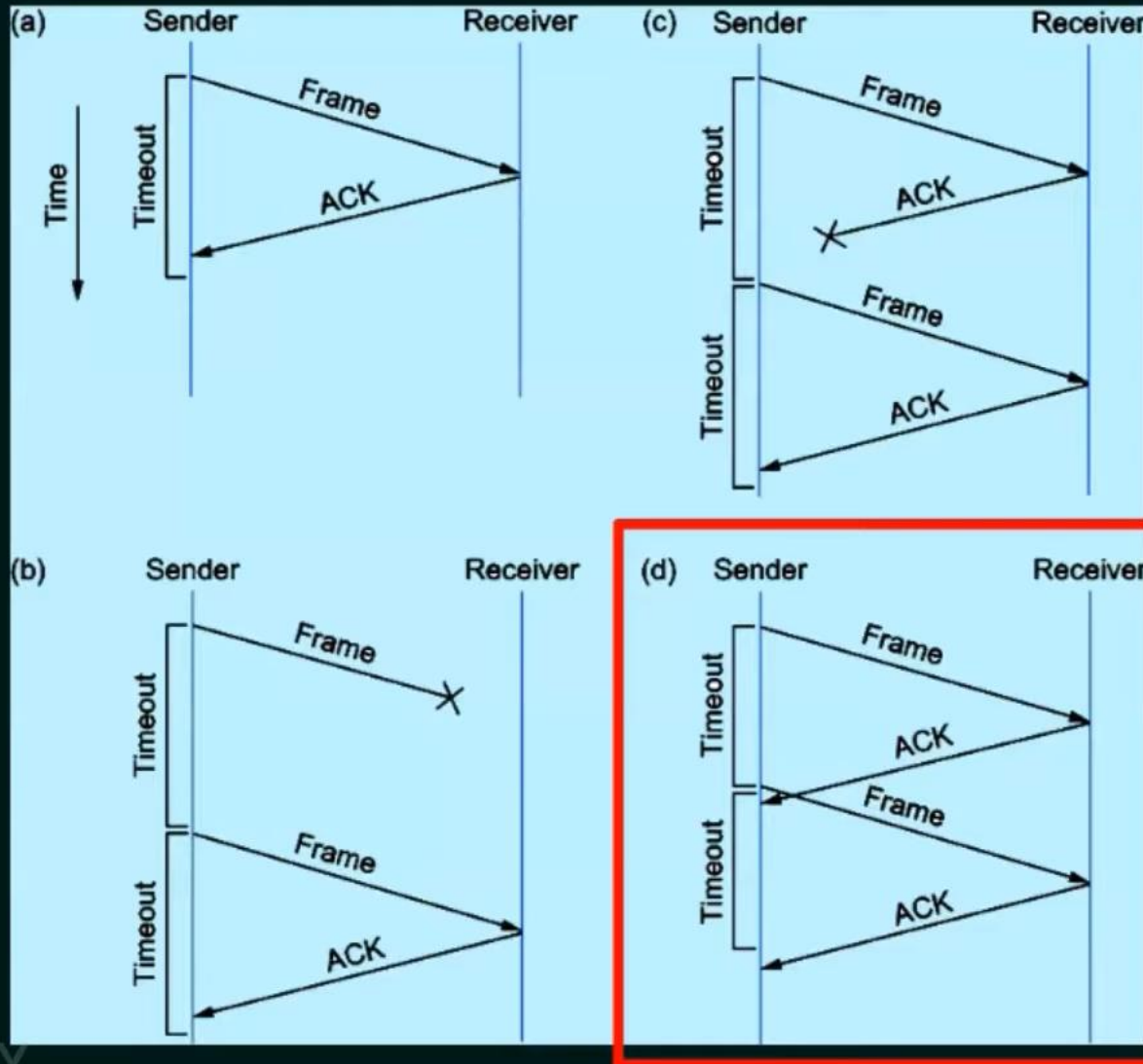
STOP-AND-WAIT ARQ PROTOCOL



(c) The ACK is lost



STOP-AND-WAIT ARQ PROTOCOL



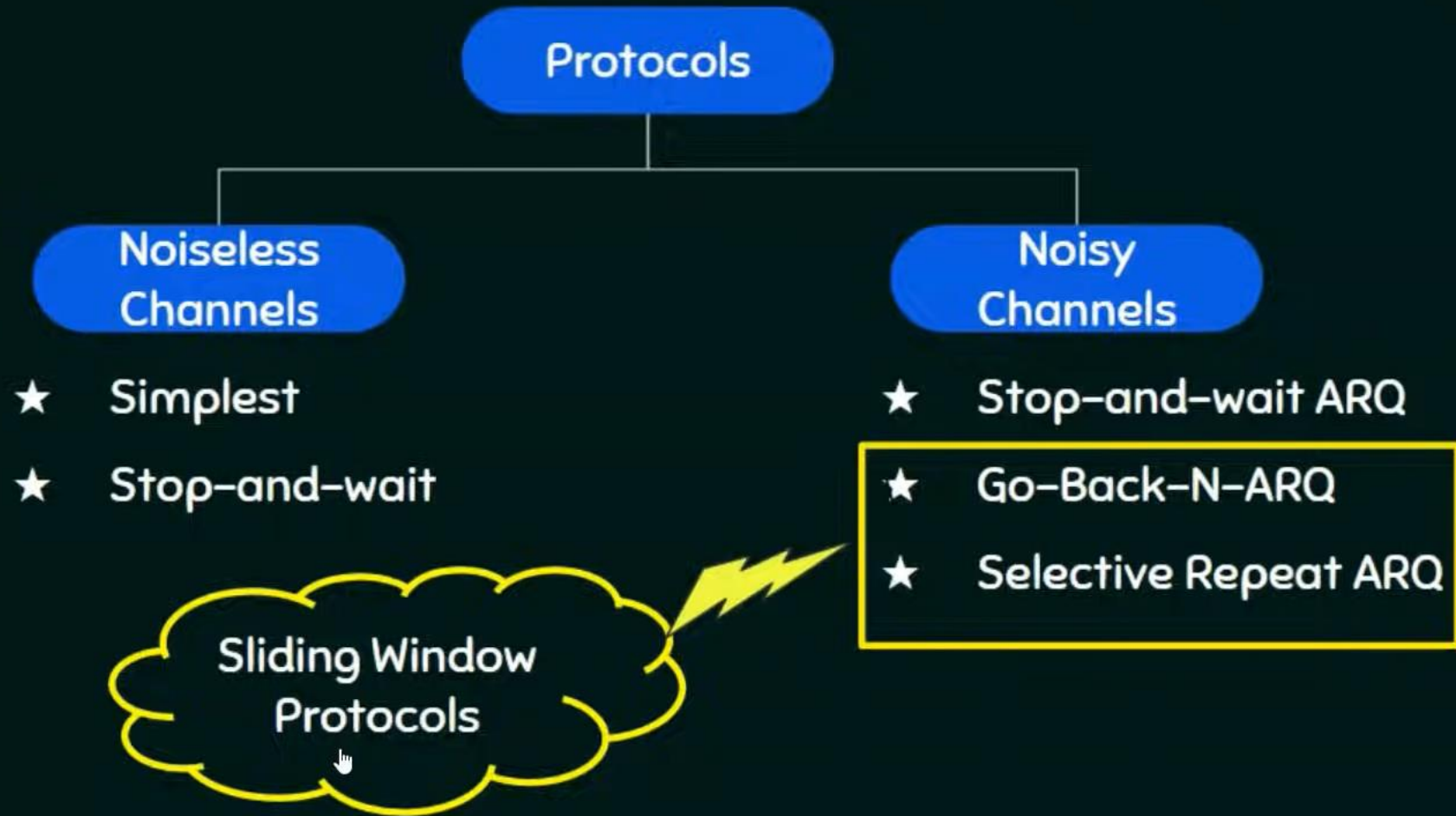
(d) The timeout fires too soon



STOP-AND-WAIT ARQ – DRAWBACKS

- ★ One frame at a time.
- ★ Poor utilization of bandwidth.
- ★ Poor Performance

FLOW CONTROL



SLIDING WINDOW PROTOCOL

- ★ Send multiple frames at a time.
- ★ Number of frames to be sent is based on **Window size**.
- ★ Each frame is numbered → Sequence number.



WORKING OF SLIDING WINDOW PROTOCOL



Window Size:

4



WORKING OF SLIDING WINDOW PROTOCOL



Window Size:

4



WORKING OF SLIDING WINDOW PROTOCOL



Window Size:

4

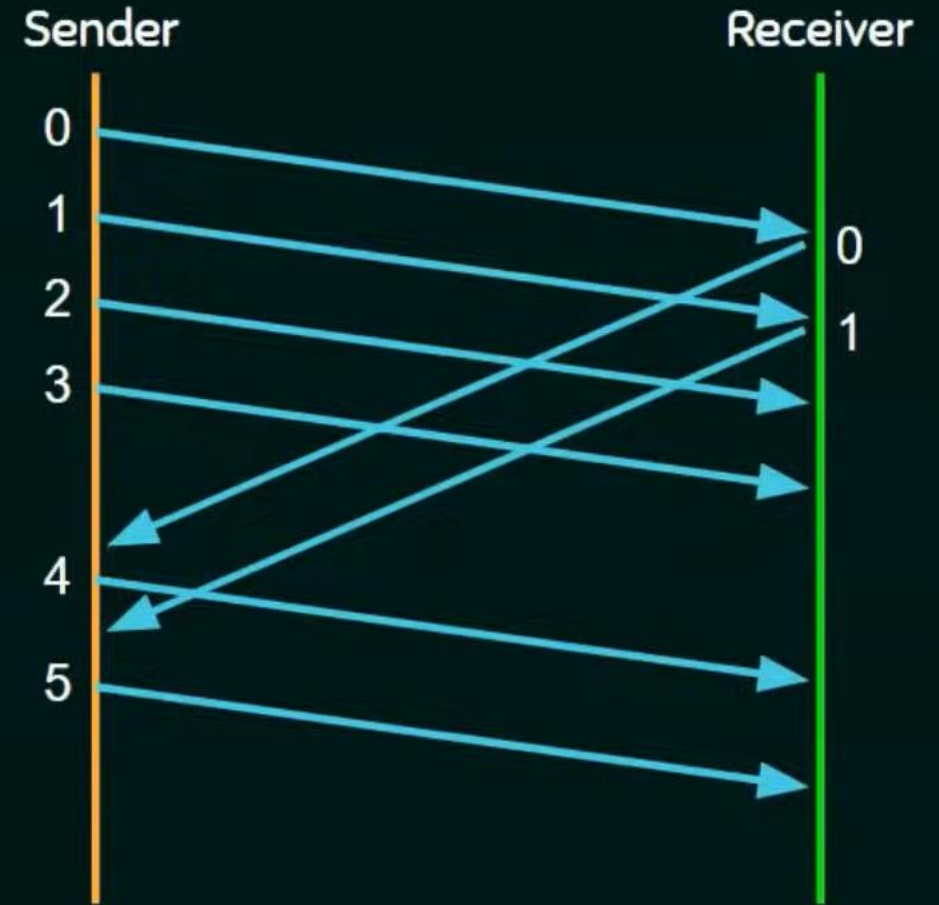


WORKING OF SLIDING WINDOW PROTOCOL



Window Size:

4



WORKING OF SLIDING WINDOW PROTOCOL

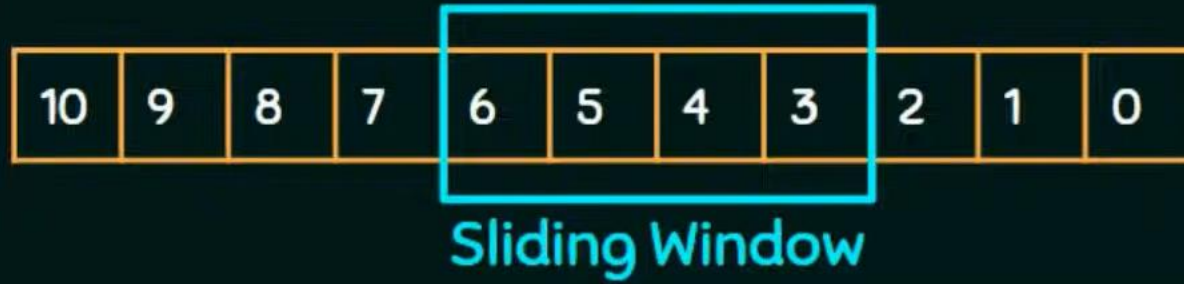


Window Size:

4

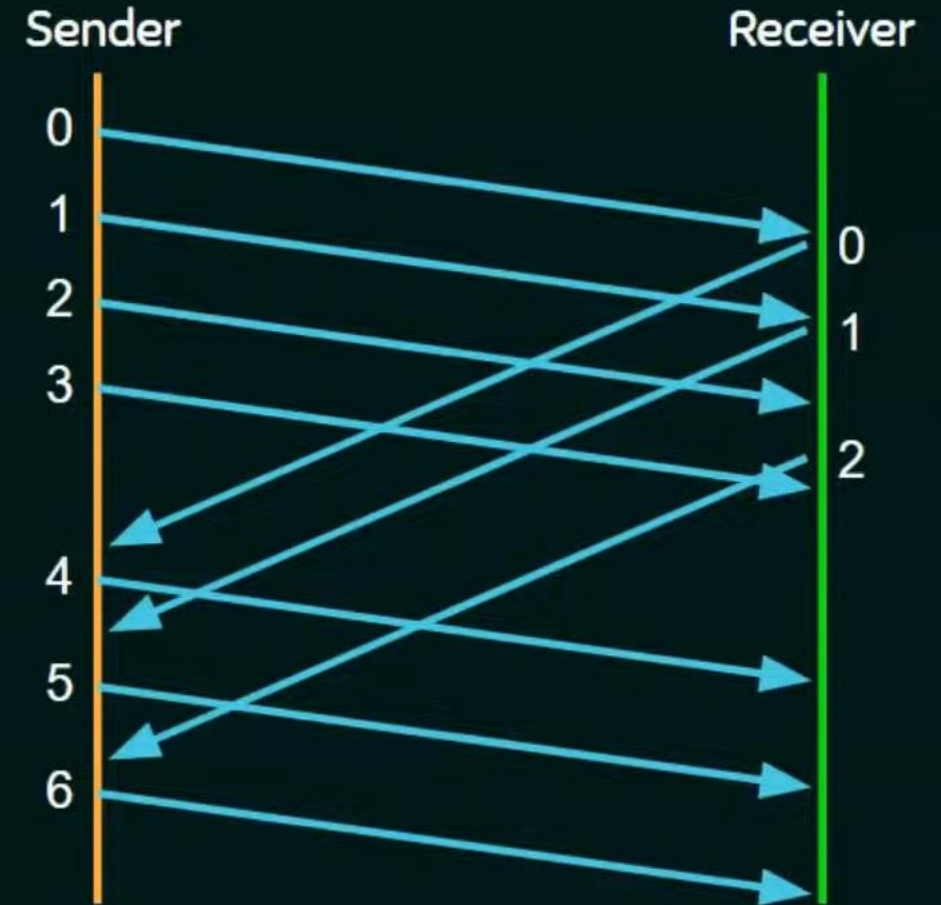


WORKING OF SLIDING WINDOW PROTOCOL

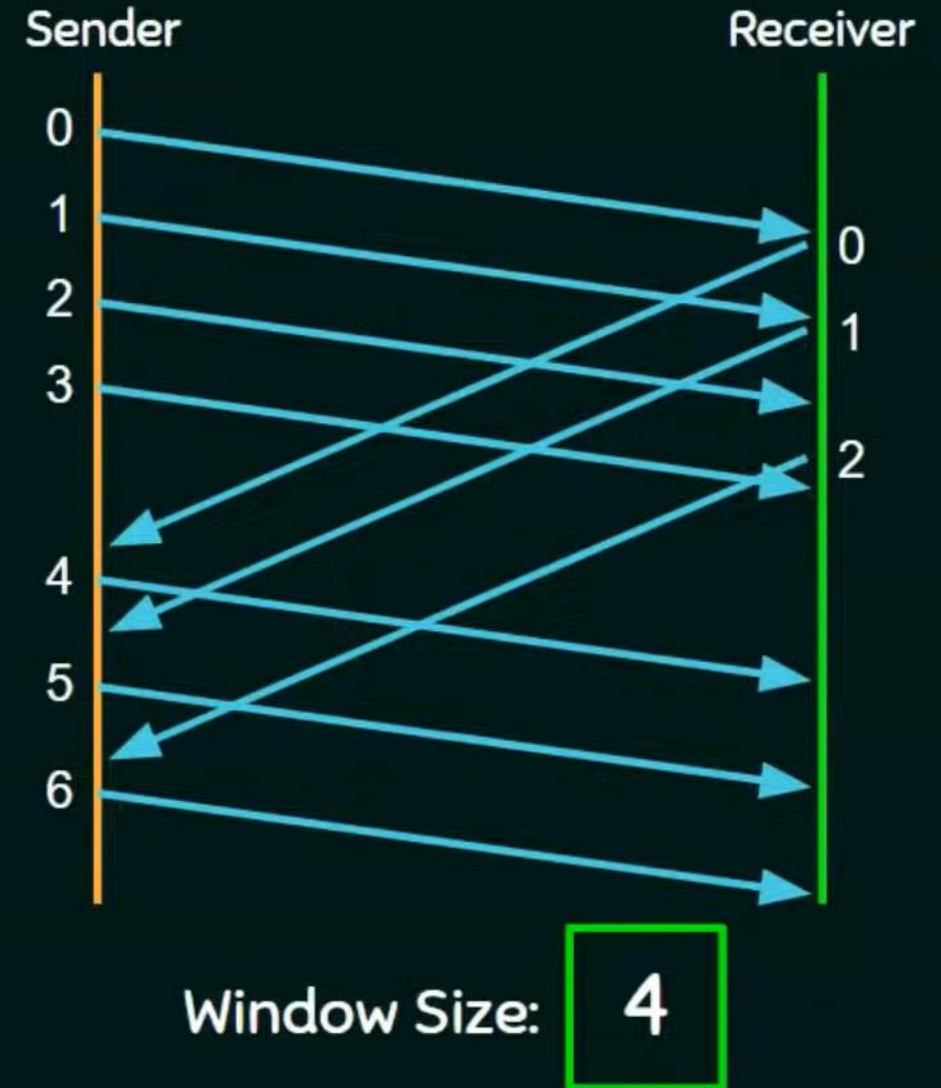
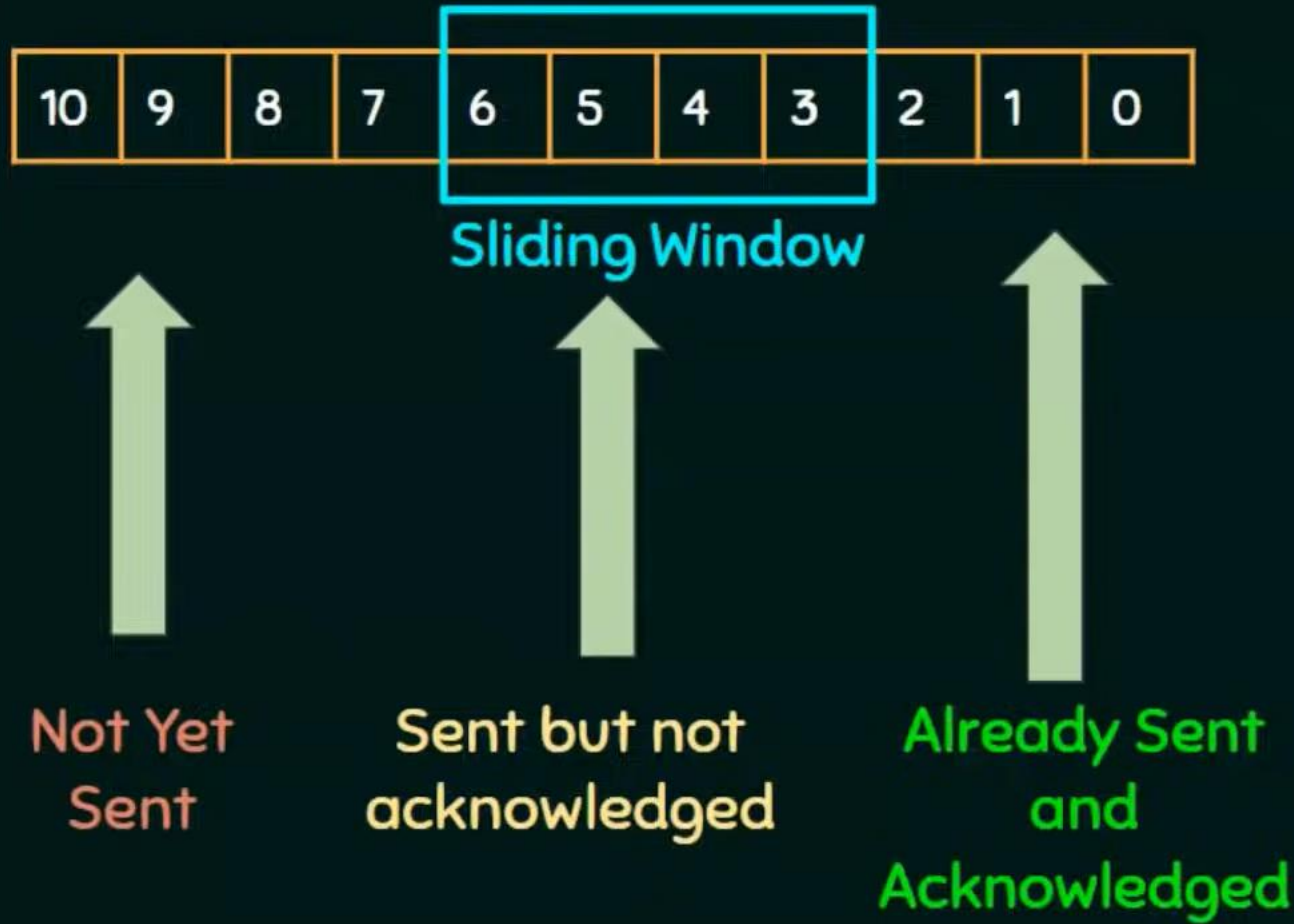


Window Size:

4



WORKING OF SLIDING WINDOW PROTOCOL



Go-Back-N ARQ



'N' is the sender window size.

Go-Back-N ARQ

- ★ Go – Back – N ARQ uses the concept of protocol pipelining i.e. the sender can send multiple frames before receiving the acknowledgment for the first frame.
- ★ There are finite number of frames and the frames are numbered in a sequential manner.
- ★ The number of frames that can be sent depends on the window size of the sender.
- ★ If the acknowledgment of a frame is not received within an agreed upon time period, all frames in the current window are transmitted.



Go-Back-N ARQ

- ★ The size of the sending window determines the sequence number of the outbound frames.



Go-Back-N ARQ

- ★ N – Sender's Window Size.
- ★ For example, if the sending window size is 4 (2^2), then the sequence numbers will be 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, and so on.
- ★ The number of bits in the sequence number is 2 to generate the binary sequence 00, 01, 10, 11.

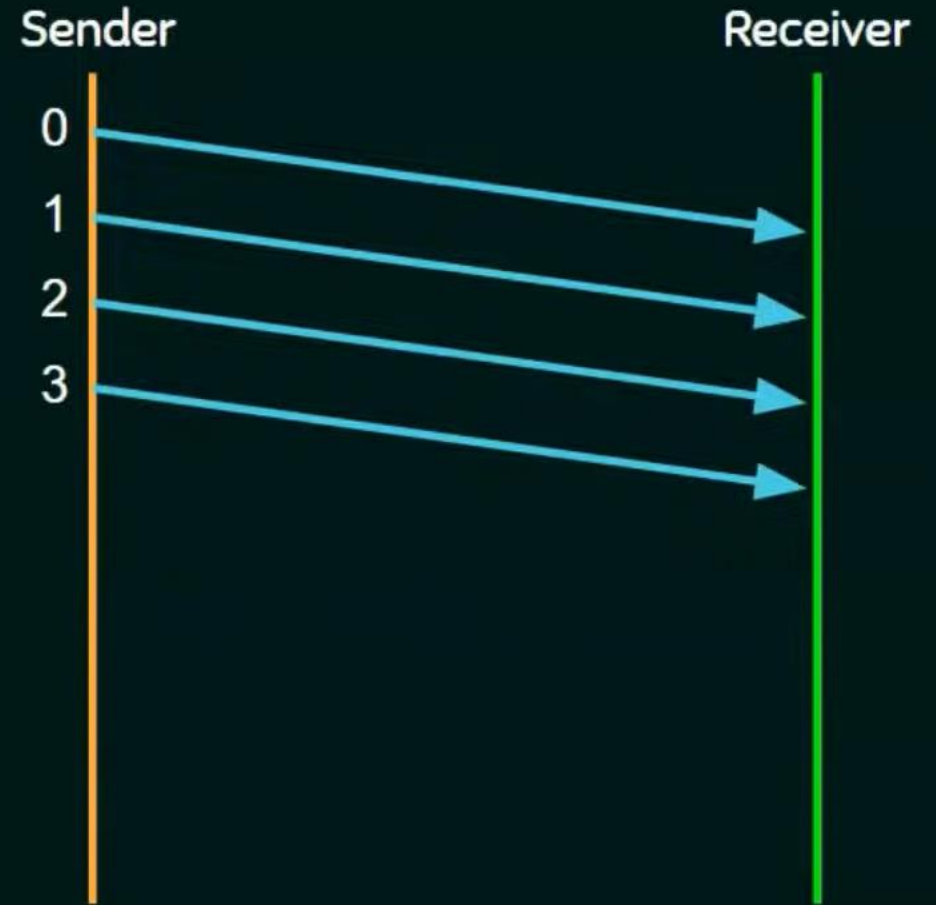


WORKING OF GO-BACK-N ARQ



Window Size:

4



WORKING OF GO-BACK-N ARQ



Window Size: 4

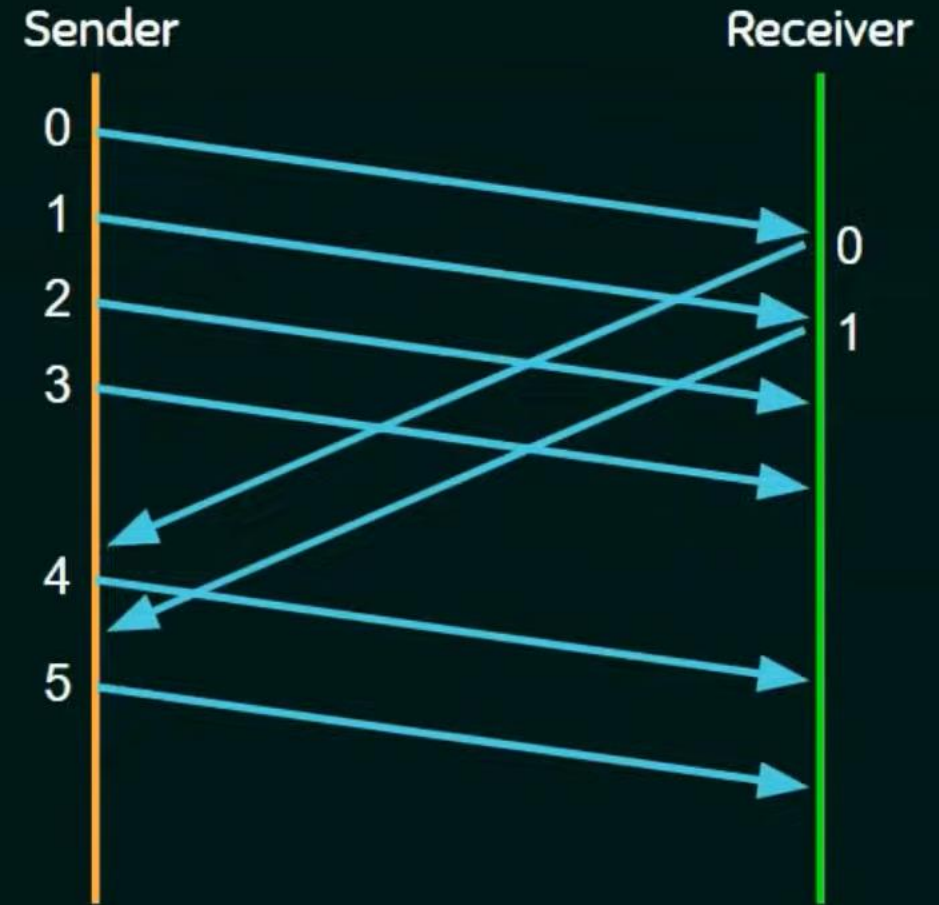


WORKING OF GO-BACK-N ARQ



Window Size:

4

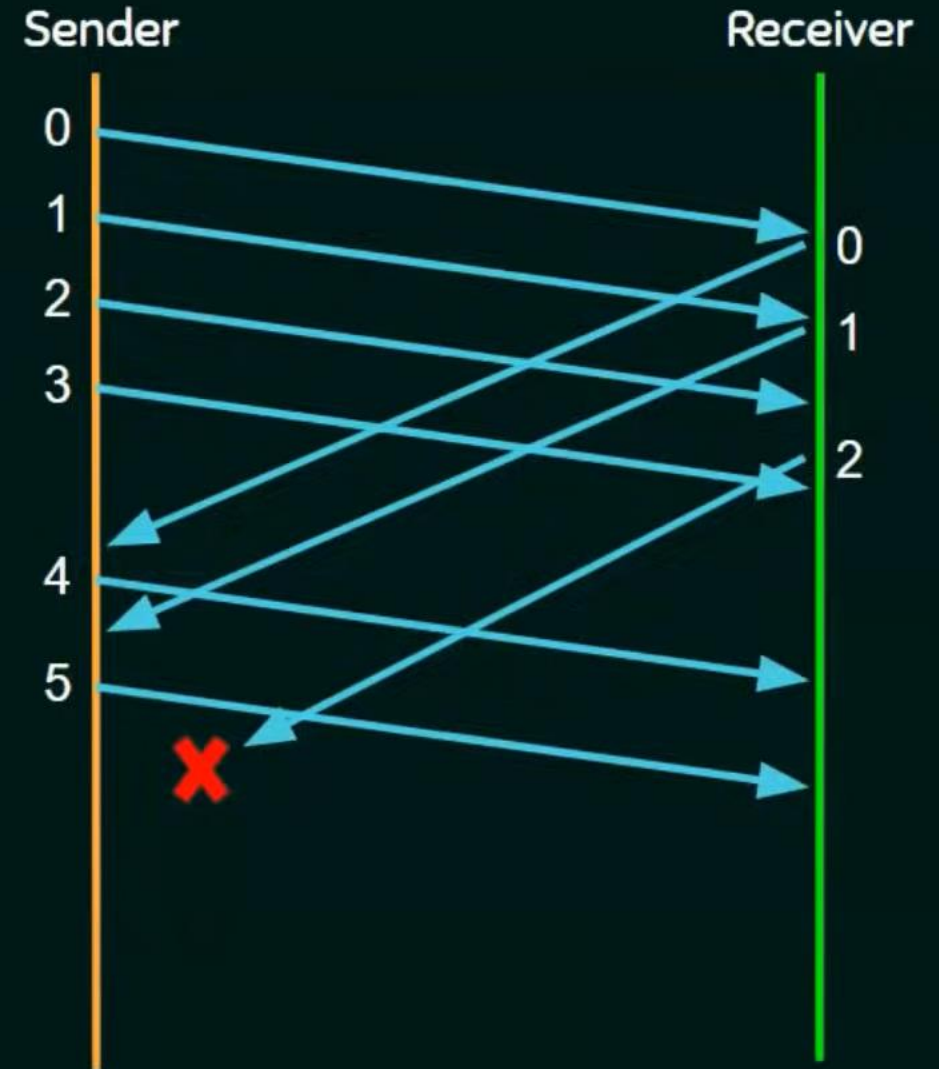


WORKING OF GO-BACK-N ARQ



Window Size: 4

ACK not received in time. So
Sender times out.

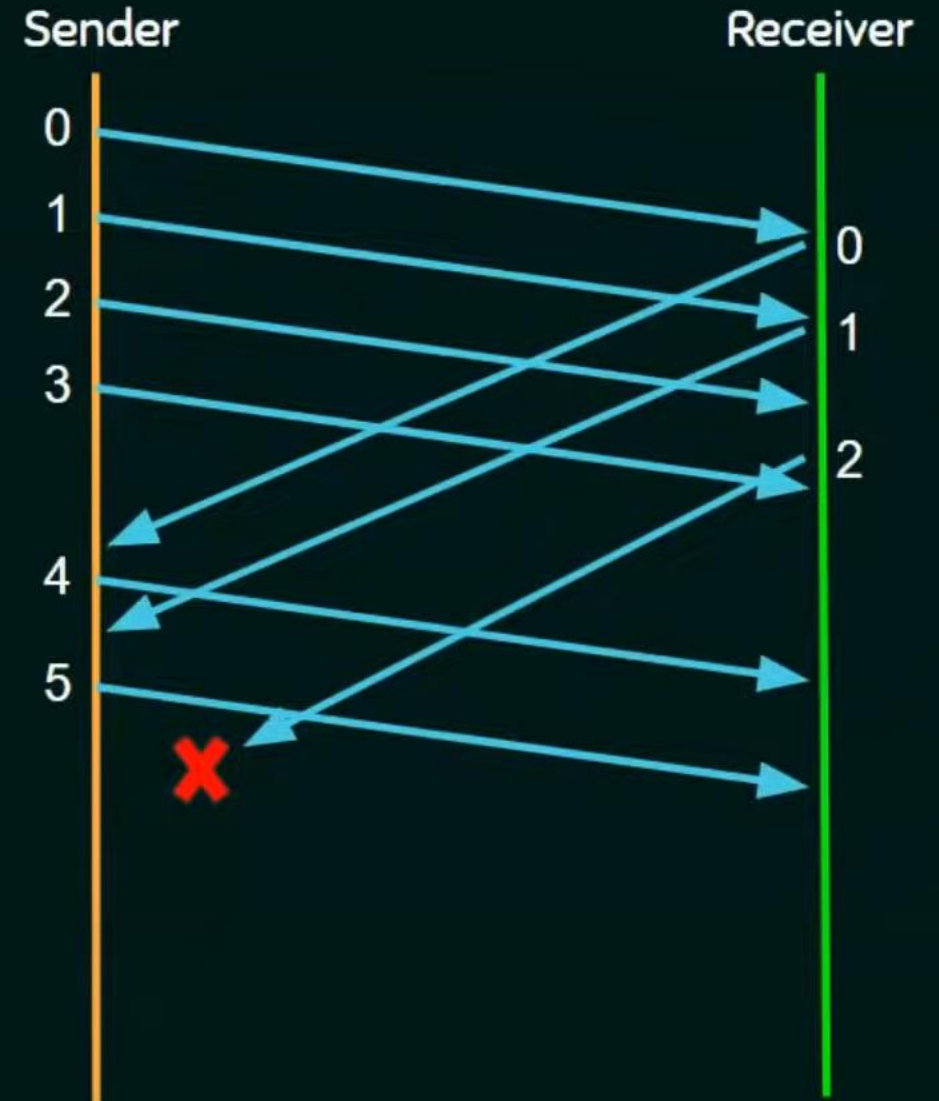


WORKING OF GO-BACK-N ARQ

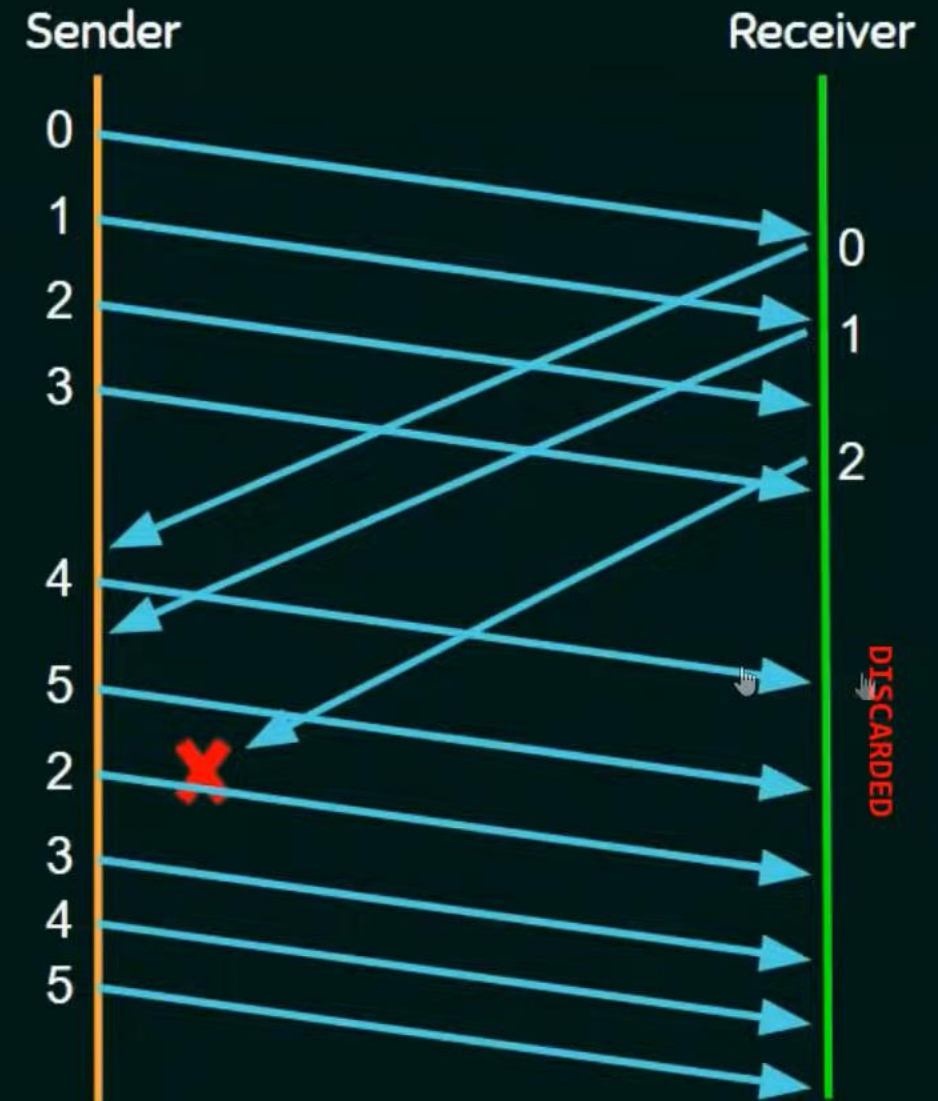
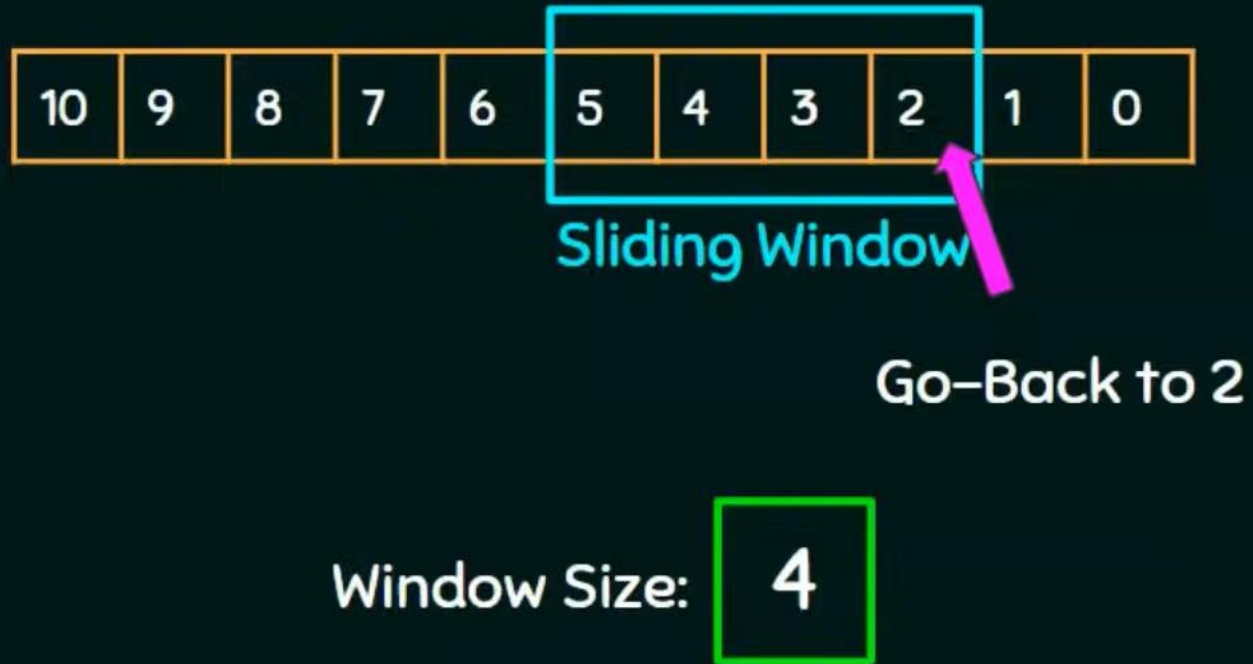


Window Size:

4



WORKING OF GO-BACK-N ARQ



QUESTION

Station A needs to send a message consisting of 9 packets to station B using a sliding window (window size 3) and go-back-n error control strategy. All packets are ready and immediately available for transmission. If every 5th packet that A transmits gets lost (but no ACKs from B ever get lost), then what is the number of packets that A will transmit for sending the message to B?

SOLUTION

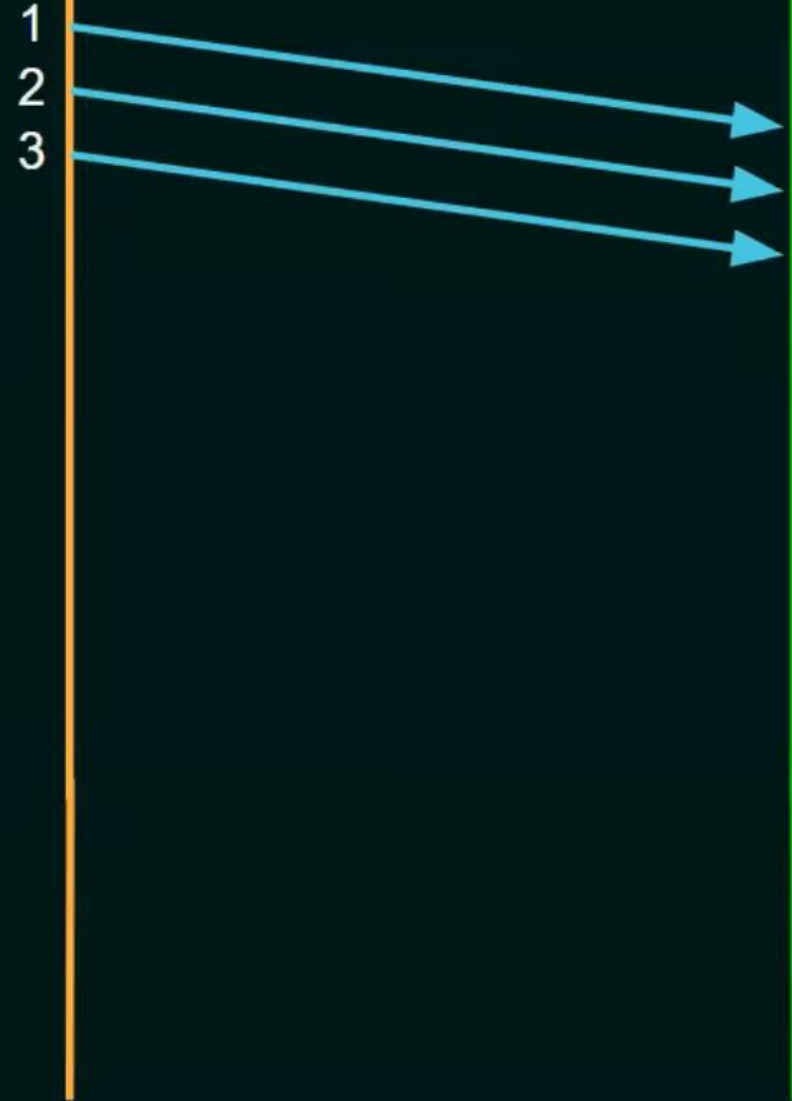


Window Size:

3

Sender

Receiver



SOLUTION

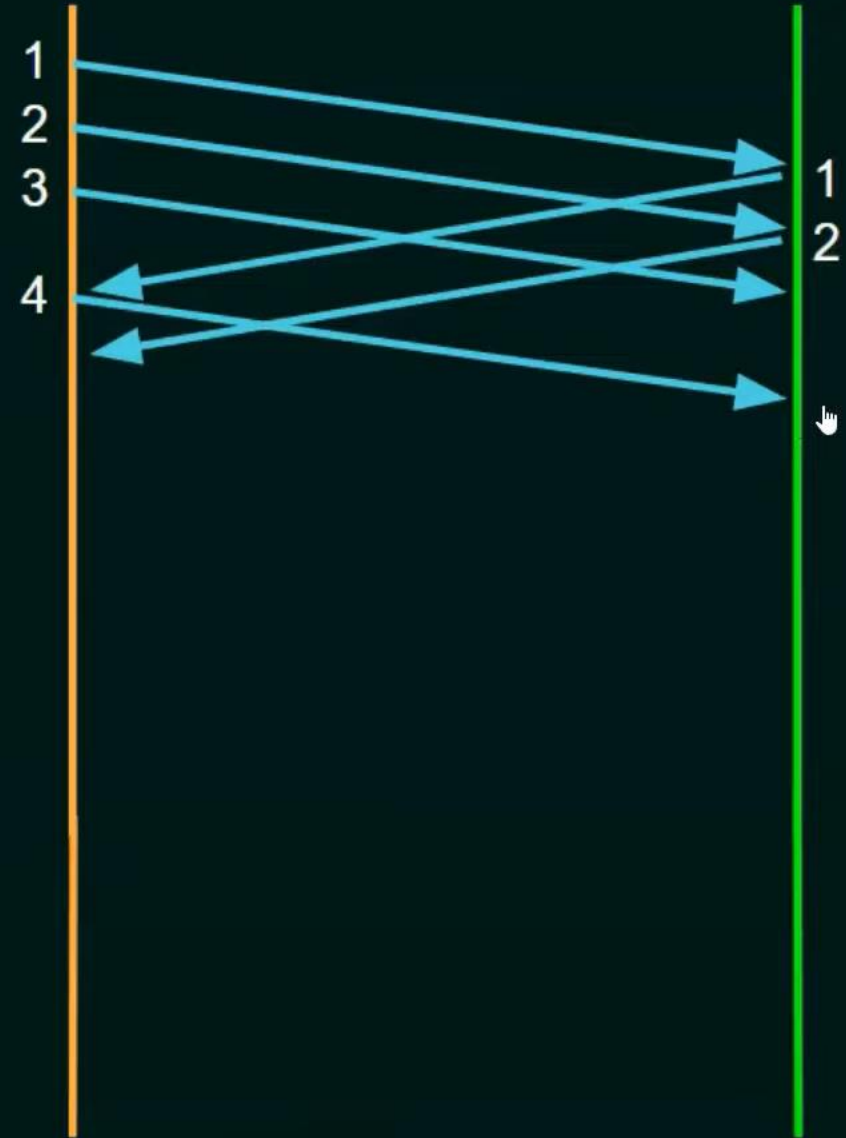


Window Size:

3

Sender

Receiver



SOLUTION

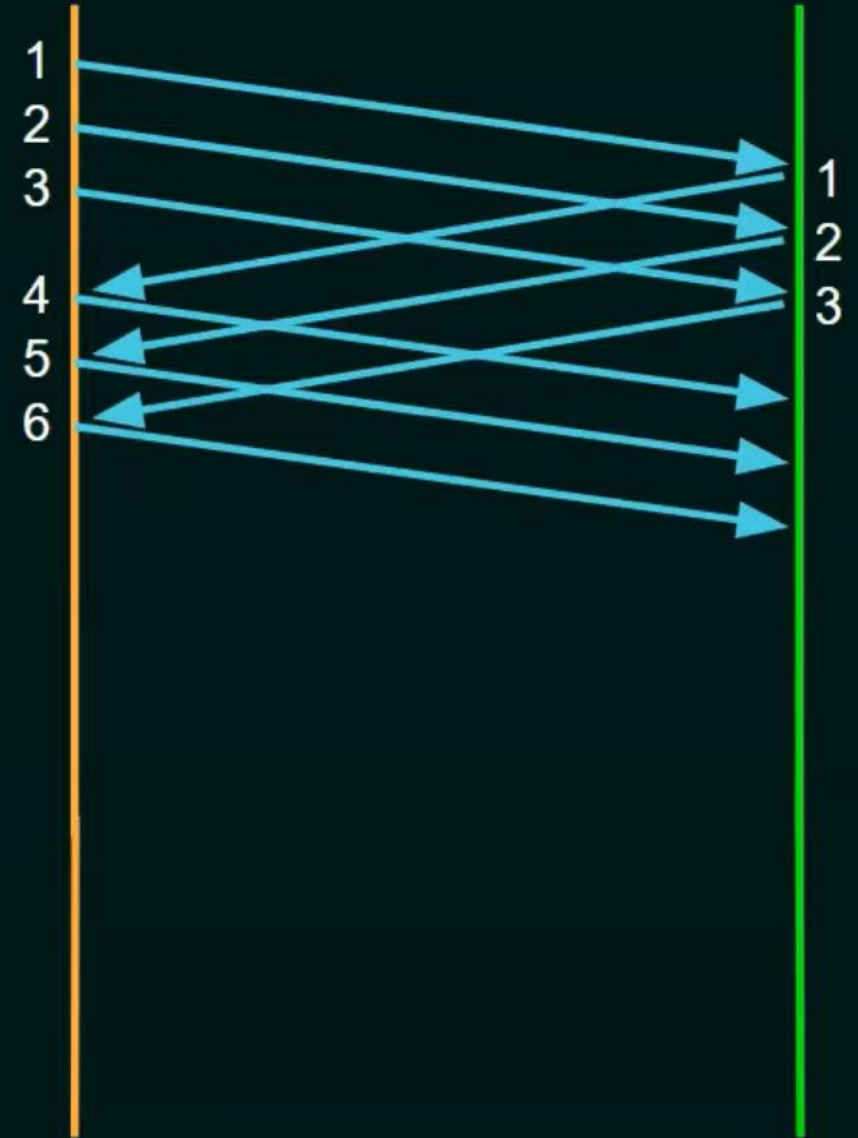


Window Size:

3

Sender

Receiver



SOLUTION

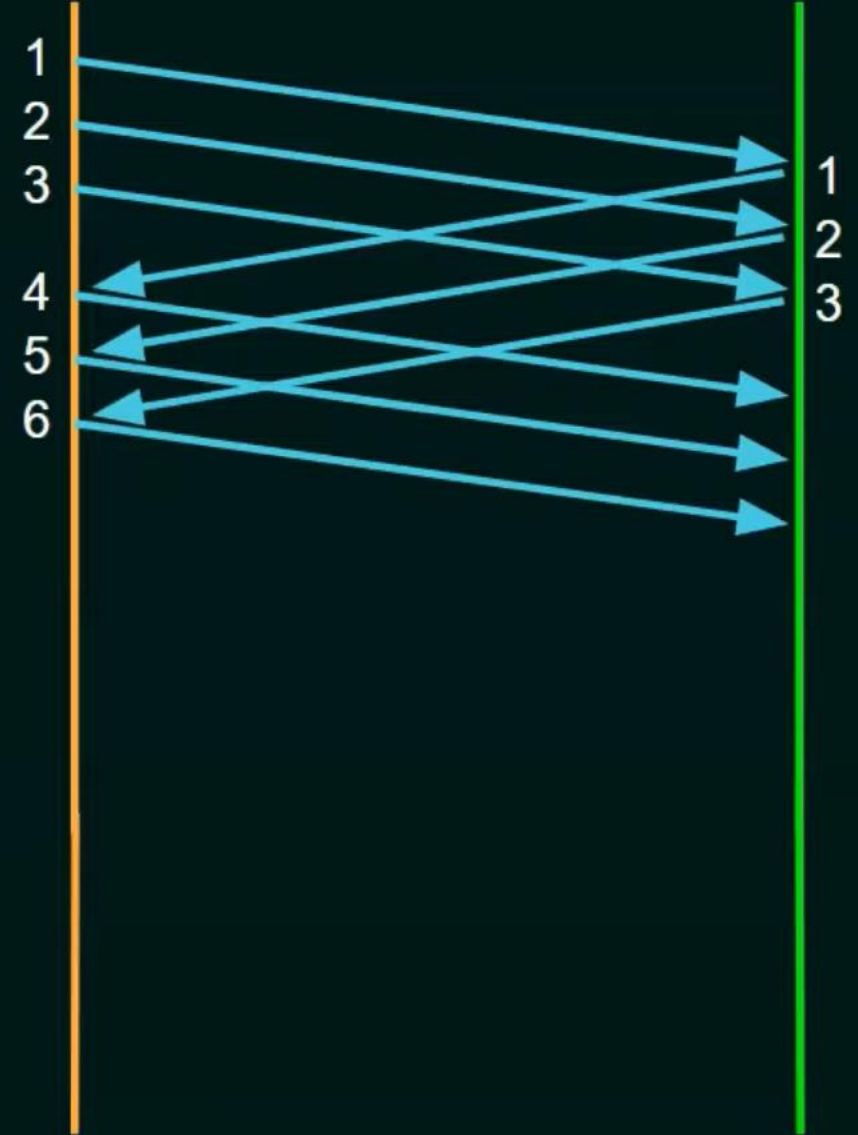


Window Size:

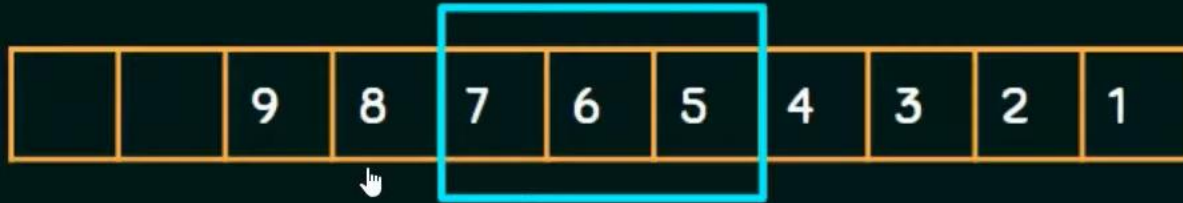
3

Sender

Receiver



SOLUTION



Window Size:

3

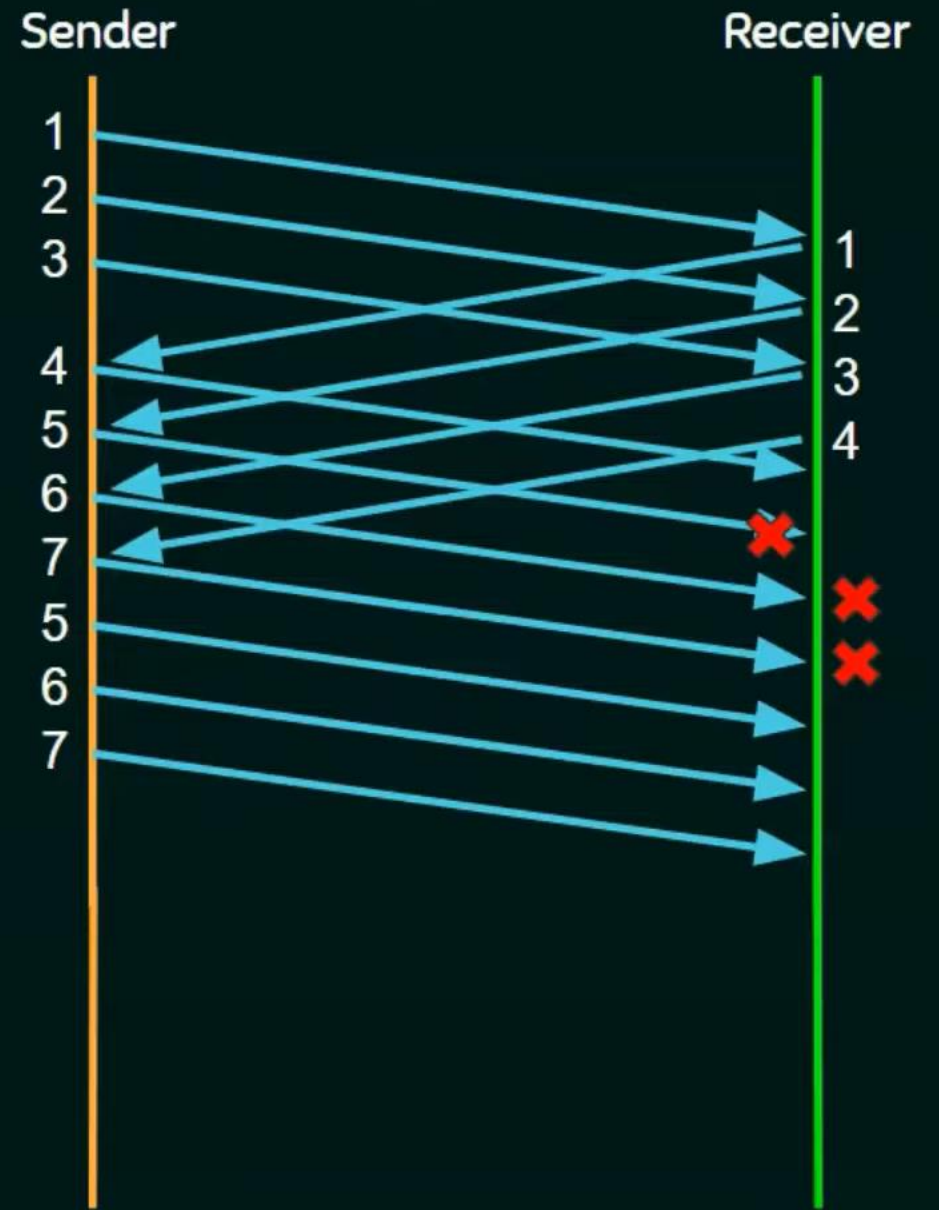


SOLUTION

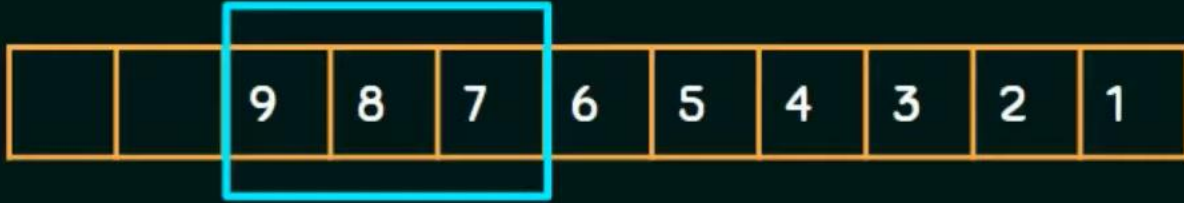


Window Size:

3



SOLUTION

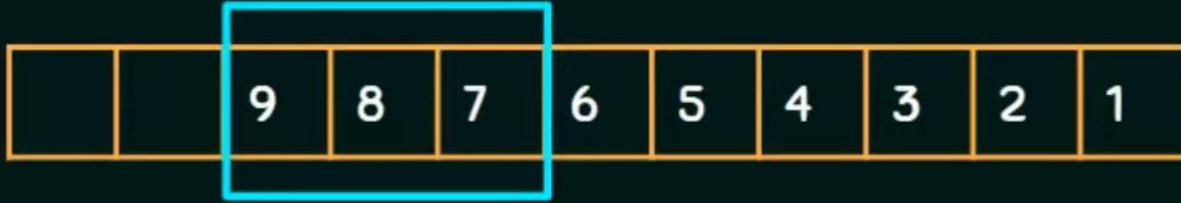


Window Size:

3

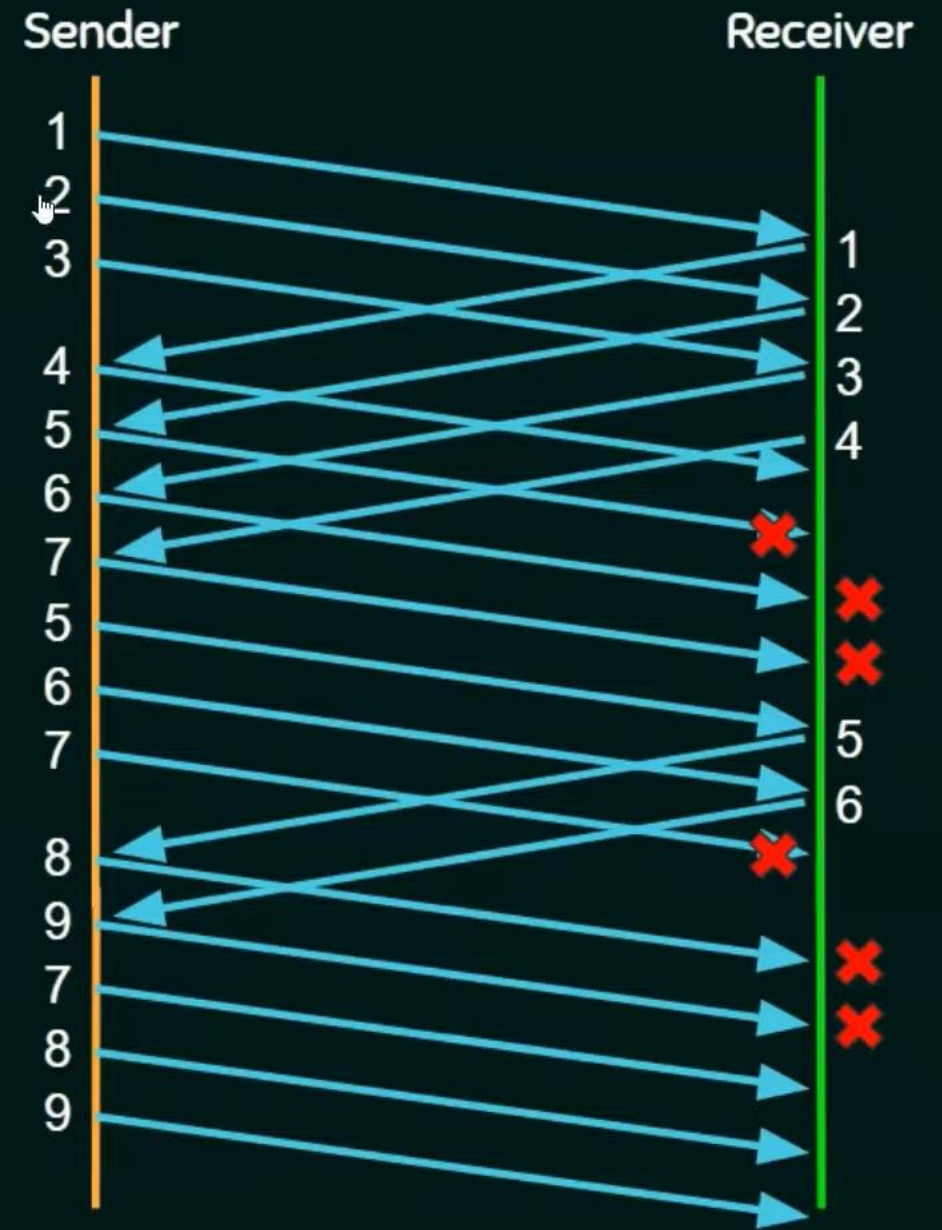


SOLUTION

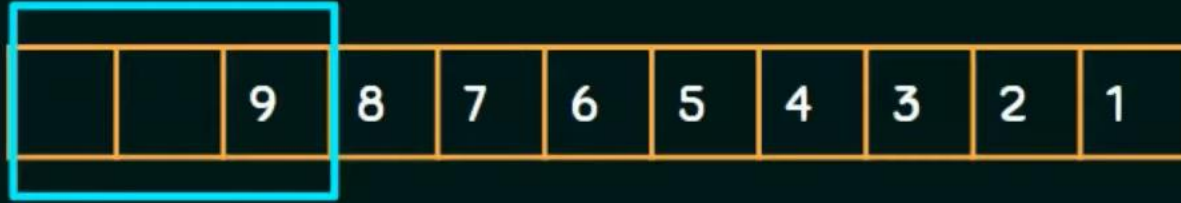


Window Size:

3



SOLUTION

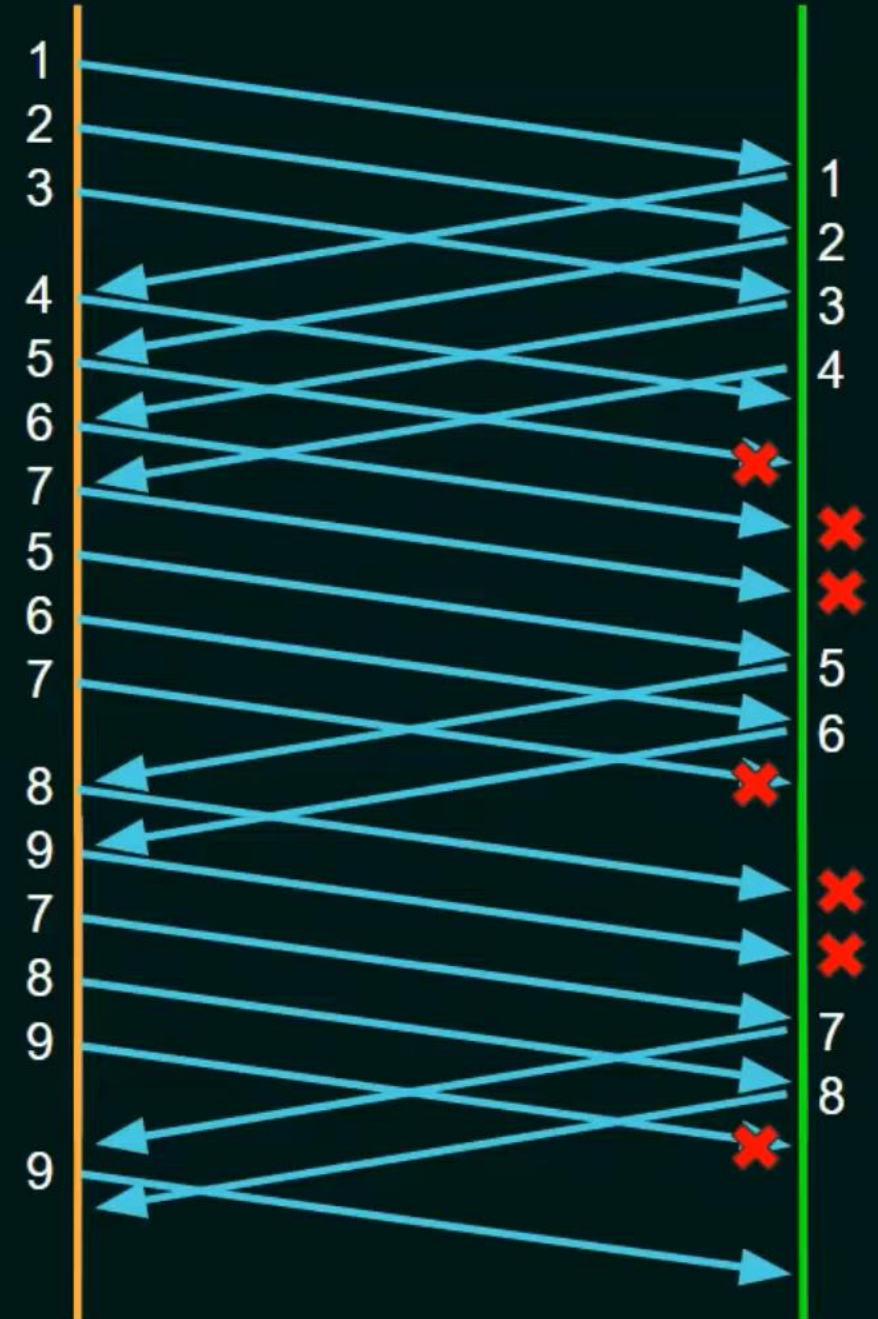


Window Size:

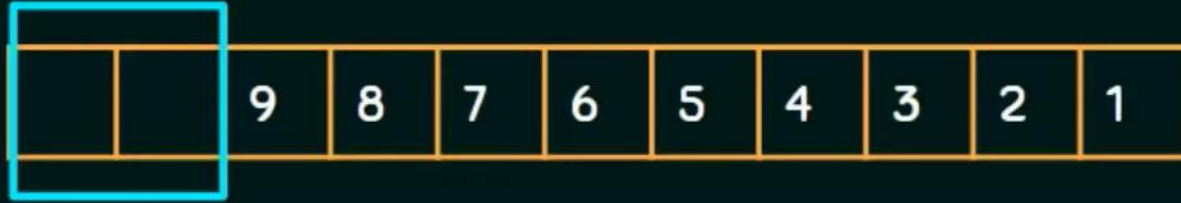
3

Sender

Receiver



SOLUTION

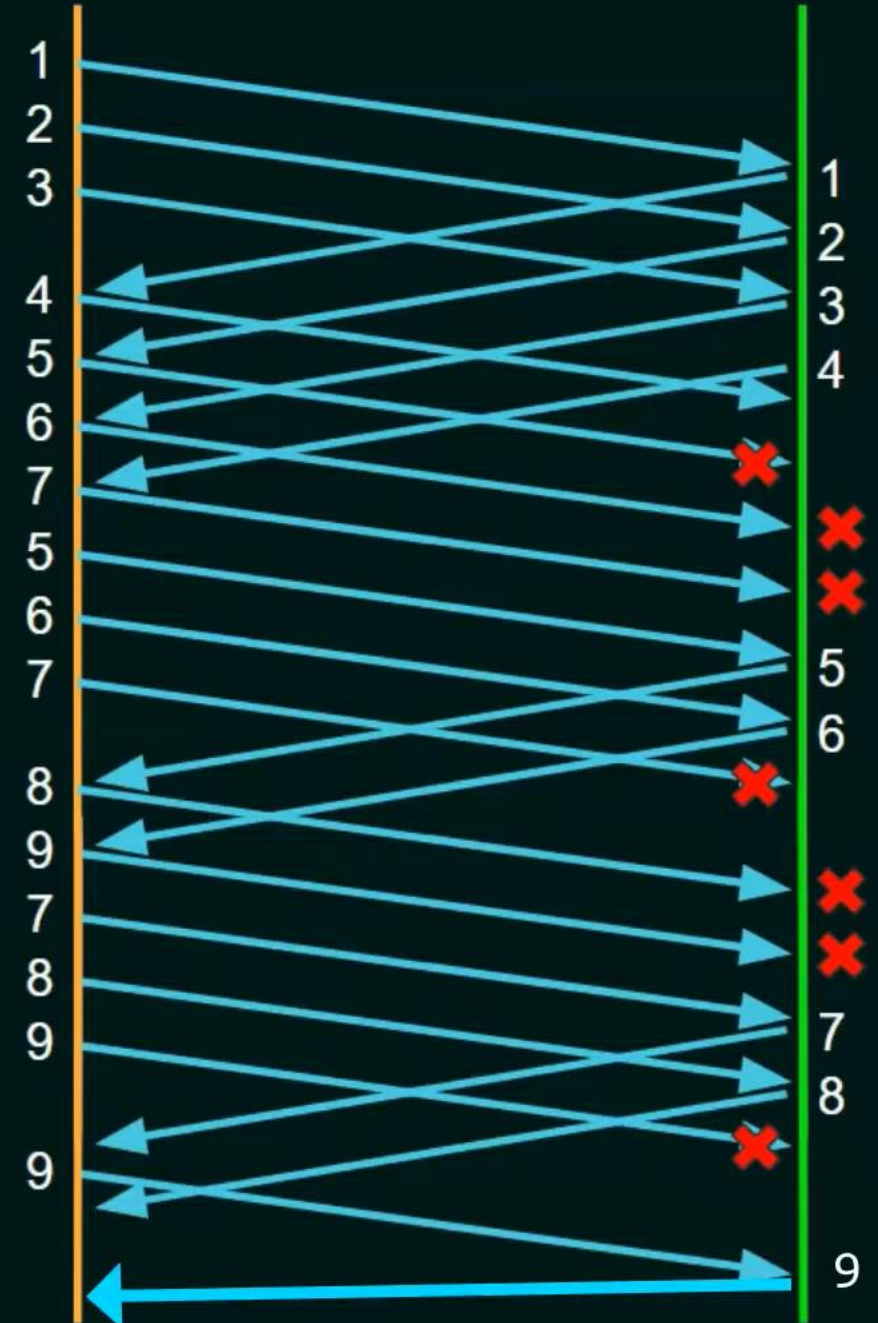


Window Size:

3

Sender

Receiver



SOLUTION



Window Size:

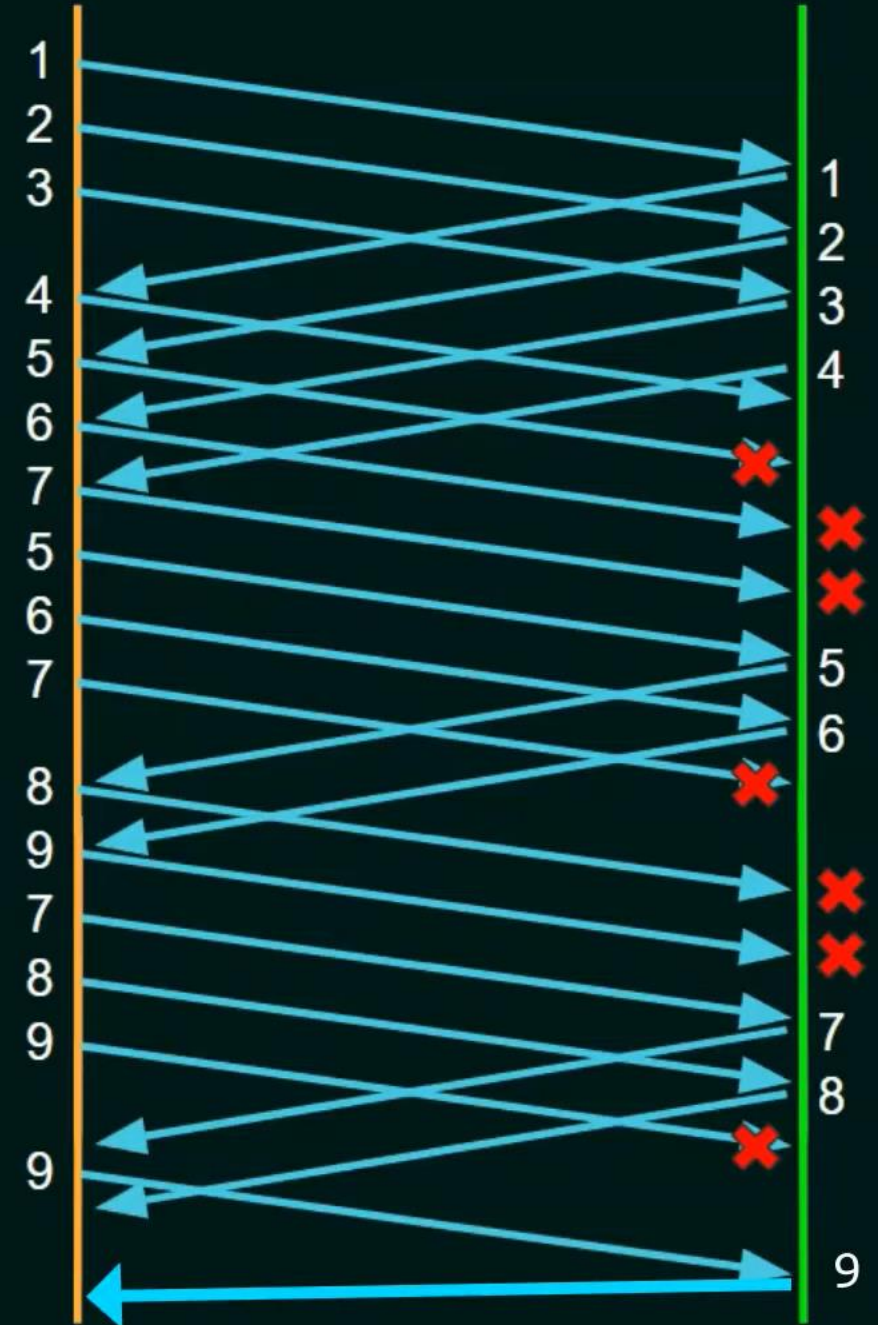
3

No. of packets transmitted by A (sender)

16

Sender

Receiver



QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with Go-Back-4. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost?

SOLUTION

Sender Window



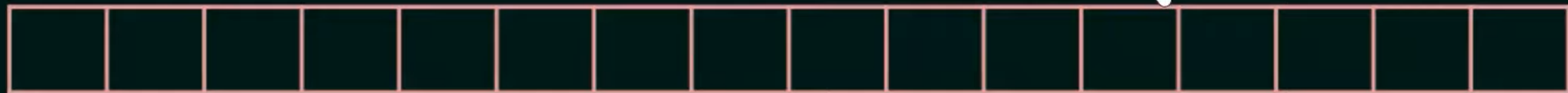
Window Size:



Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window

10	9	8	7	6	5	4	3	2	1
----	---	---	---	---	---	---	---	---	---

Window Size:

4

Frames transmitted by Host A

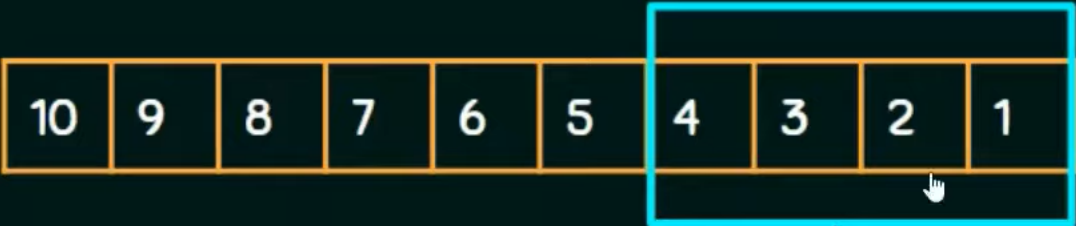
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Frames Acknowledged by Host B

--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

SOLUTION

Sender Window



Window Size:



Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window



Window Size:



Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window



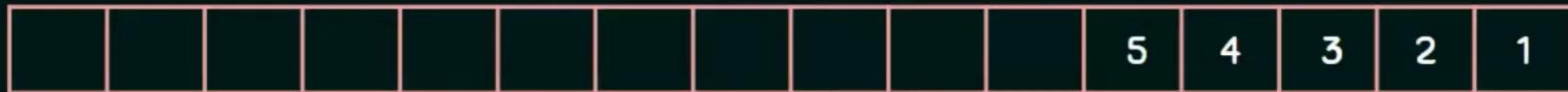
Window Size:

4

Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window



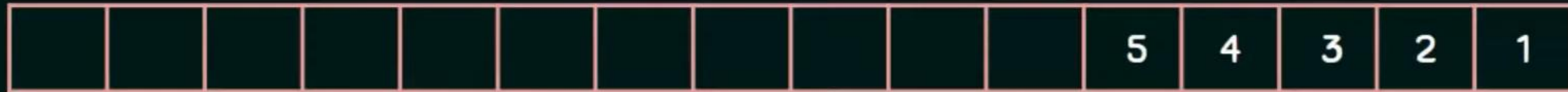
Window Size:

4

Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window



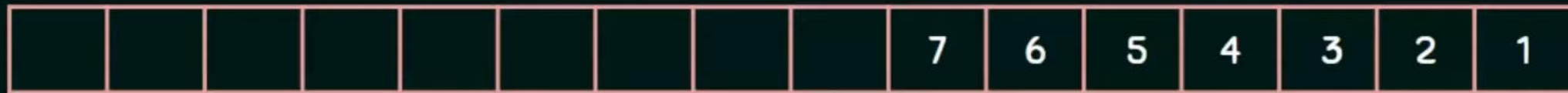
Window Size:



Frames transmitted by Host A



Frames Acknowledged by Host B



SOLUTION

Sender Window



Window Size:

4

Frames transmitted by Host A



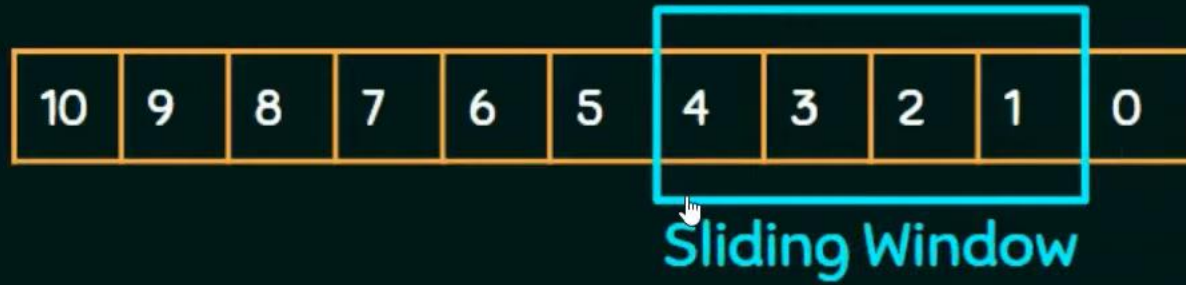
Frames Acknowledged by Host B



SELECTIVE REPEAT ARQ

- ★ In Selective Repeat ARQ, only the erroneous or lost frames are retransmitted, while correct frames are received and buffered.
- ★ The receiver while keeping track of sequence numbers, buffers the frames in memory and sends NACK for only frame which is missing or damaged.
- ★ The sender will send/retransmit packet for which NACK is received.

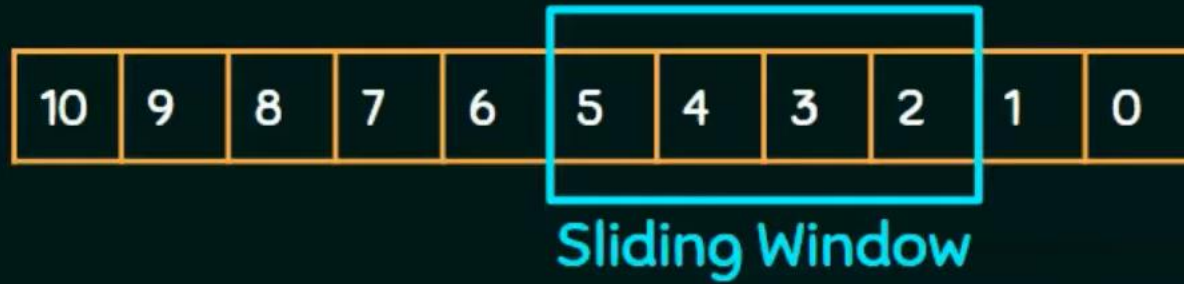
WORKING OF SELECTIVE REPEAT



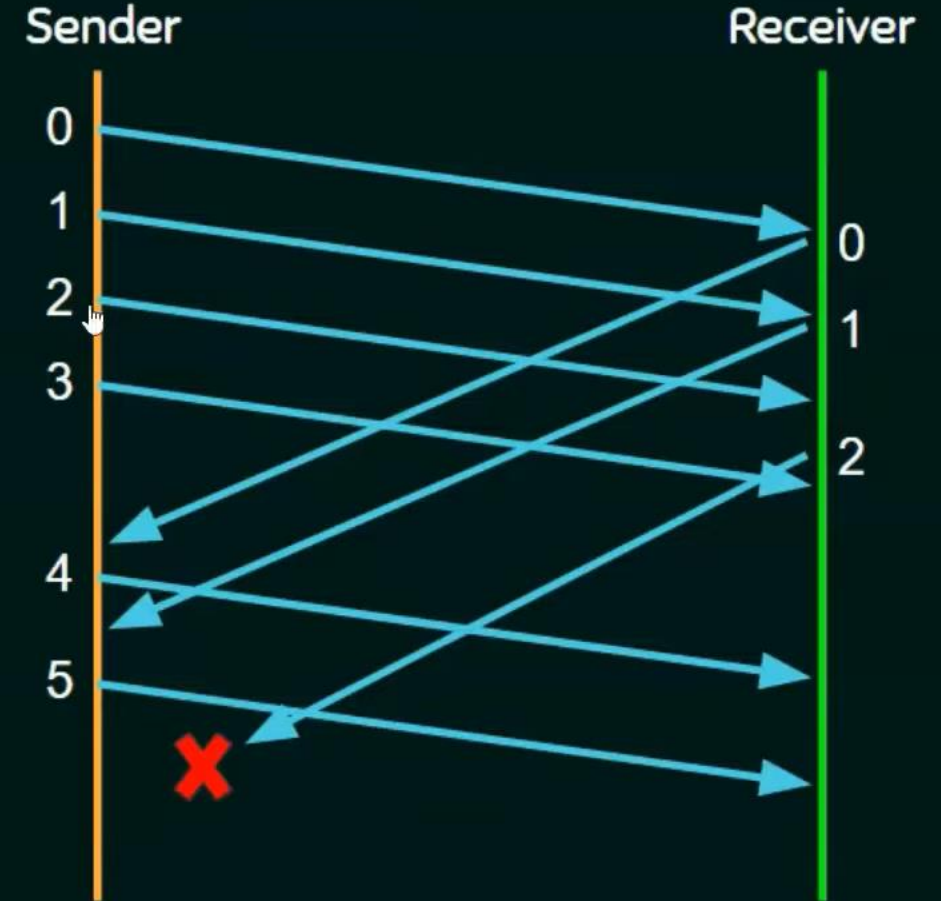
Window Size: 4



WORKING OF SELECTIVE REPEAT



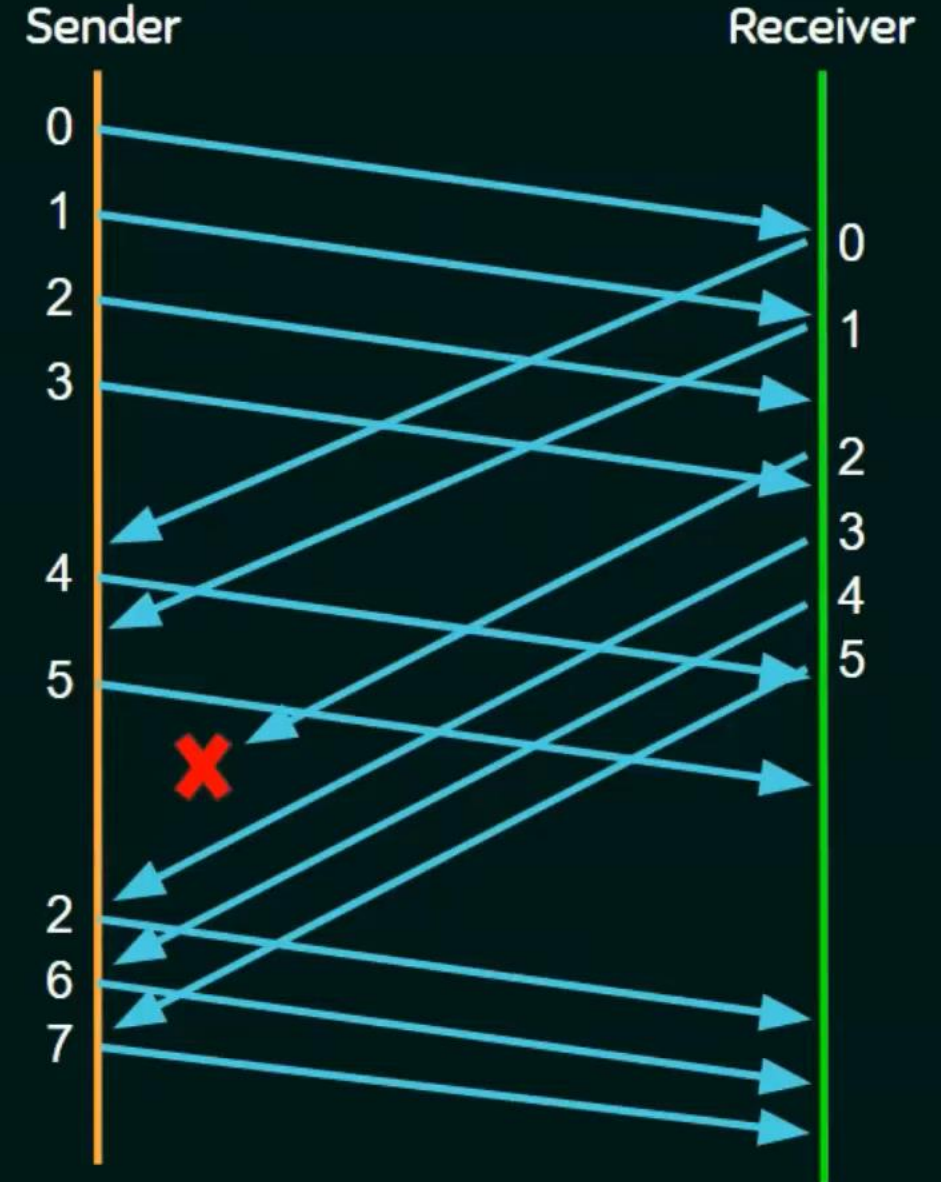
Window Size: 4



WORKING OF SELECTIVE REPEAT



Window Size: 4



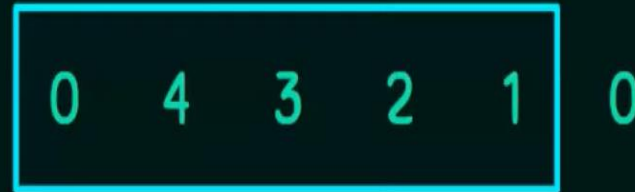
QUESTION

In SR protocol, suppose frames through 0 to 4 have been transmitted. Now, imagine that 0 times out, 5 (a new frame) is transmitted, 1 times out, 2 times out and 6 (another new frame) is transmitted. At this point, what will be the outstanding packets in sender's window?

Sender's Window



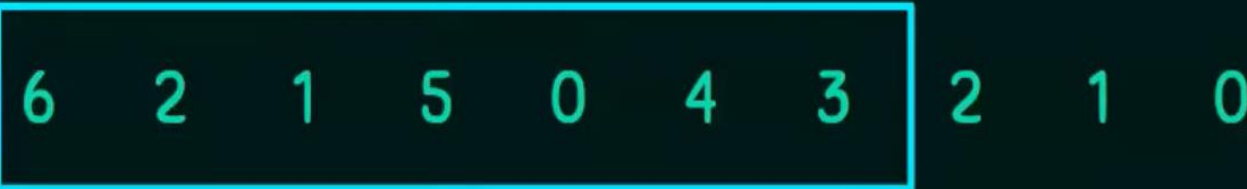
Sender's Window



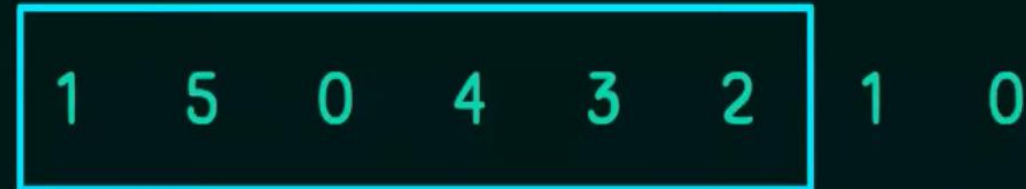
Sender's Window



Sender's Window



Sender's Window



QUESTION

Host A wants to send 10 frames to Host B. The hosts agreed to go with SR ARQ. How many number of frames are transmitted by Host A if every 6th frame that is transmitted by host A is either corrupted or lost? Also compare the number of transmissions of SR ARQ with Go-Back-4 ARQ.

SOLUTION

Number of transmission with Go-Back-4 ARQ: 17

Refer previous lecture titled: Go-Back-N ARQ (Solved Problem 2)

10	9	8	7	6	5	4	3	2	1
----	---	---	---	---	---	---	---	---	---

Frames transmitted by Host A

												10	9	6	8	7	6	5	4	3	2	1
--	--	--	--	--	--	--	--	--	--	--	--	----	---	---	---	---	---	---	---	---	---	---

Number of frames transmitted by Host A: 11

ANSWER

Go-Back-N: Number of frames transmitted by Host A: 17

Selective Repeat: Number of frames transmitted by Host A: 11